MCOD-FM-RobustBench

Foundation-Model Promptability and Spectral-Degradation Robustness Benchmark for Multispectral Camouflaged Object Detection

Title

MCOD-FM-RobustBench: Foundation-Model Promptability and Spectral-Degradation Robustness Benchmark for Multispectral Camouflaged Object Detection

Primary Goal

Build a scientifically defensible benchmark-extension paper on the official **MCOD** dataset that answers the following question:

> How reliably do general-purpose segmentation foundation models identify camouflaged objects from multispectral imagery, and how does their performance change when spectral evidence or spectral alignment is degraded?

Main Task Type

Binary camouflaged object mask prediction / promptable segmentation on 8-band multispectral images.

Primary Dataset

MCOD: The First Challenging Benchmark for Multispectral Camouflaged Object Detection - ACM Multimedia 2025.

The official MCOD paper already:

· introduced the 8-band MCOD dataset;

· reported 1,527 registered and calibrated images with pixel masks and eight challenge attributes;

· established a 1,027 training / 500 testing split;

· benchmarked eleven COD models on MCOD;

· compared RGB and multispectral inputs for selected COD models;

· analysed attributes such as small objects, illumination, background similarity and complex backgrounds.

HyperCOD/HSC-SAM (AAAI 2026) already adapts SAM to hyperspectral camouflage using spectral-spatial decomposition and adaptive spectral prompting.

Therefore, it can be positioned as:

An MCOD-specific foundation-model promptability and spectral-degradation robustness benchmark, with optional lightweight reference baselines.

Project Execution Overview

Stage	Work to Complete	Main Output	Stop/Continue Rule
1	Download MCOD and official resources	Raw dataset folder and source log	Stop if archive is unavailable or licence constraints are ignored
2	Inspect raw files and official split	Band/mask/attribute verification report	Stop if masks or bands cannot be matched
3	Organise data and create manifests	Standardised local folder structure	Continue only after leakage checks
4	Run complete dataset audit	Spectral statistics and visualisations	Required before inference
5	Reproduce benchmark anchors	One or two RGB/MSI COD model outputs	Establish pipeline correctness only
6	Run SAM and SAM2 zero-shot promptability benchmark	Prompt protocol table	Central new benchmark stage
7	Evaluate fair predicted-prompt pipelines	RGB-prompt versus MSI-prompt results	Tests whether spectral evidence helps promptability
8	Run spectral view and band-group analysis	Per-band/group table	New MCOD-FM analysis
9	Run spectral degradation robustness benchmark	Missing-band/misalignment/corruption tables	Central new benchmark stage
10	Run attribute-wise and failure analysis	Failure taxonomy and figures	Required for paper story
11	Evaluate efficiency and reproducibility	FPS, latency, parameters, seeds	Required for WACV-quality evaluation
12	Optional reference baseline / strong extension	Only after novelty review	Do not turn into an HSC-SAM duplicate

TASK 1: Download the Required Datasets and Resources

1.1 Primary Dataset - Download First

Dataset / Resource	Official or Author-Linked Link	Verified Details	Use in Project	Priority
MCOD official repository	https://github.com/yl2900260-bit/MCOD	ACM MM 2025 repository; includes licence, band definitions, dataset links and author contact	Primary authoritative source	Required
MCOD Google Drive dataset folder	Google Drive dataset folder	Linked directly from official repository; link resolved during verification	Download main dataset	Required
MCOD BaiduNetdisk alternative	BaiduNetdisk link	Linked from official repository; extraction/password code 1234	Alternative if Drive fails	Optional alternative
Official MCOD reported results archive	Google Drive MSIResults.zip	Linked from official repository	Compare reported baseline maps/results;	Required reference
MCOD ACM paper	https://doi.org/10.1145/3746027.3758294	Published at ACM Multimedia 2025	Required reading and citation	Required
MCOD arXiv text	https://arxiv.org/abs/2509.15753	Public manuscript with details on data, split, attributes and experiments	Working reference	Required

1.2 MCOD Verified Dataset Facts

Item	Verified Information
Modality	Multispectral imagery, 8 channels
Number of images	1,527
Train/test split	1,027 training images / 500 testing images
Annotation type	Pixel-level ground-truth masks and challenge attribute labels
Image resolution reported	1140 x 860
Attribute count	Eight attributes

Object size range	0.01% to 4.16% of image area
Average object area	0.429% of image area
Official paper existing experiments	Eleven COD methods; selected RGB versus MSI comparison; attribute analysis
Dataset licence	CC BY-NC-ND 4.0; academic research only; no modification or redistribution without permission
Dataset contact listed by repository	yl2900260@gmail.com

1.3 MCOD Spectral Bands

Band	Start Wavelength (nm)	Stop Wavelength (nm)	Central Wavelength (nm)	Colour / Region	Use in our Analysis
S1	395	450	422.5	Violet	Optional visible-band analysis
S2	455	520	487.5	Blue	Official false-colour view component
S3	525	575	550.0	Green	Official false-colour view component
S4	580	625	602.5	Orange	Optional visible-band analysis
S5	630	690	660.0	Red	Official false-colour view component
S6	705	745	725.0	Red Edge	Required spectral-view analysis
S7	750	820	785.0	NIR 1	Required NIR analysis
S8	825	950	887.5	NIR 2	Required NIR analysis

Important: The MCOD paper states that calibrated bands S2, S3 and S5 were used to reconstruct false-colour composites. We must inspect the downloaded format and document the exact channel stacking convention used locally before treating it as an RGB-like image.

1.4 Required Auxiliary RGB COD Datasets - Download Only After MCOD Works

These datasets are used only to validate RGB baseline code or to confirm that a COD model implementation is functioning before it is adapted to MCOD. They are not central to the new contribution.

Dataset	Official / Author-Linked Link	Verified Details	Use	Priority
COD10K	Official COD project page / Official download link from that page	10,000 images; official page reports 6,000 training and 4,000 test images, with 5,066 camouflaged images in total	Check RGB COD model implementation and report standard reference performance only if needed	Recommended auxiliary
CAMO	Official CAMO project page	1,250 camouflaged images: 1,000 training and 250 testing; objectness masks	RGB baseline validation	Optional auxiliary
NC4K	Official release repository	4,121 camouflaged test images with ground-truth and instance-level annotation	Test-only RGB generalisation validation	Optional auxiliary
CHAMELEON	Available through official SINet testing-data distribution: SINet repo	76-image legacy COD testing dataset	Small sanity-check test only	Optional

TASK 2: Inspect Raw MCOD Dataset Before Preparing Anything

5.1 Keep the Raw Dataset Unchanged

Place the original downloaded dataset archive and extracted dataset here:

```
data/raw/MCOD/
```

Never rename or overwrite original data files. Any normalisation, visual composite or corrupted image must be written under data/processed/ or local temporary output folders.

5.3 Required Checks

Check	What the Script Must Confirm	Output File
Archive/file readability	All files can be opened	raw_corrupted_files.csv
Total sample count	Verify 1,527 samples	raw_sample_count.csv
Official split	Verify 1,027 train / 500 test if split files/folders are included	official_split_verification.csv
Eight-band completeness	Every sample has S1-S8 or equivalent 8-channel file	missing_band_report.csv
Band shape consistency	All bands for one sample have identical height/width	band_shape_report.csv
Mask match	Each image has a corresponding binary mask	missing_mask_report.csv
Attribute labels	Attribute representation exists and can be parsed	attribute_format_report.txt
File data type	Confirm uint8, uint16 or float and scaling	dtype_dynamic_range_report.csv
Composite view availability	Identify whether an official RGB/false-colour composite is provided	composite_view_report.txt
Duplicate/leakage check	Hash and perceptual-hash duplicate checks across train/test	train_test_leakage_report.csv

TASK 6: Prepare Standardised MCOD Views and Manifests

6.2 Standardised Data Views

We must create local derived views for experiments but must not redistribute those derived images.

View Name	Bands / Data Used	How It Is Constructed	Experiment Purpose
official_false_colour	S2, S3, S5	Follow paper/archive convention; document channel order after verification	
S6_rededge_gray3	S6	Normalised S6 repeated into 3 channels	
S7_nir1_gray3	S7	Normalised S7 repeated into 3 channels	

S8_nir2_gray3	S8	Normalised S8 repeated into 3 channels	Test NIR2 visibility
visible_group_projection	S1-S5	Normalise and project locally; rule documented	Visible-only group analysis
nir_group_projection	S6-S8	Normalise and project locally; rule documented	Spectral group analysis
all8_input	S1-S8	Native tensor only for COD baseline/model stages, not directly for SAM/SAM2 unless projected	MSI COD baseline / prompt-source analysis
ground_truth_mask	Official mask	Read unchanged	Evaluation only

6.3 Manifest File

Create:

```
data/manifests/mcod_manifest.csv
```

Required columns:

```
sample_id,split,path_s1,path_s2,path_s3,path_s4,path_s5,path_s6,path_s7,path_s8,path_false_colour,path_mask,attributes,height,width,dtype,min_s1,max_s1,min_s8,max_s8,object_area_ratio,duplicate_group,notes
```

6.4 Attribute Columns

The manifest or a joined attribute table must support these official MCOD attributes:

Code	**Definition**
SC	Shape Complex
SO	Small Object: object-to-image area ratio < 0.001
OC	Object Occlusion
BS	Background Similarity
UI	Uneven Illumination
CB	Complex Background
OE	Overexposure
II	Insufficient Illumination

6.6 TASK 6 Deliverables

```
data/manifests/mcod_manifest.csv

data/reports/spectral_view_generation_log.csv

data/reports/channel_order_verification.txt

outputs/figures/first_20_band_view_grid.png
```

TASK 7: Run the Complete Dataset Audit

7.2 Audit Operations

Audit Item	Operation	Output
Sample count	Count all train/test samples	dataset_summary.csv
Mask count	Verify one mask per sample	mask_integrity_report.csv
Band count	Verify S1-S8 for every sample	band_integrity_report.csv
Resolution	Min/max/mean resolution and distribution	resolution_distribution.csv
Dtype and dynamic range	Per-band dtype, min, max, mean, std	band_statistics.csv
Object area	Mask pixels divided by image pixels	object_area_distribution.csv
Official small objects	Verify SO threshold against mask area ratio < 0.001	small_object_verification.csv
Attribute distribution	Count each official challenge attribute	attribute_distribution.csv
Attribute co-occurrence	Pairwise co-occurrence counts	attribute_cooccurrence.csv
Train/test similarity	Compare size and attribute distributions	split_distribution_check.csv
Duplicate images	Exact hash check	duplicate_exact_report.csv
Near duplicates	Perceptual hash or feature similarity screen	duplicate_near_report.csv

Sample visual inspection	Show bands, composite and GT mask	sample_band_visualization.png

7.3 Visualisations to Produce

```
outputs/figures/sample_band_visualization.png
outputs/figures/sample_false_colour_mask_overlay.png
outputs/figures/band_mean_std_plot.png
outputs/figures/object_area_histogram.png
outputs/figures/attribute_distribution_bar.png
outputs/figures/attribute_cooccurrence_heatmap.png
outputs/figures/train_test_size_distribution.png
```

7.4 Dataset Audit Table

Dataset	Modality	Channels	Samples	Train	Test	Masks	Attributes	Resolution	Licence	Student Use
MCOD	MSI	8	1,527	1,027	500	Pixel-level	8	1140 x 860 reported	CC BY-NC-ND 4.0	Primary benchmark
COD10K	RGB	3	10,000 total	6,000	4,000	Multi-annotation COD labels	Attributes/categories	Verify after optional download	Official project distribution	Auxiliary code validation only
CAMO	RGB	3	1,250 camouflaged images	1,000	250	Objectness mask	Not central	Verify after optional download	Verify from project page	Auxiliary only
NC4K	RGB	3	4,121	Test only	Test only	GT + instance annotation	Not central	Verify after optional download	Verify	Auxiliary only

TASK 8: Create the Evaluation Pipeline Before Running Models

8.1 Evaluation Tool

Use:

Tool	Official Link	Purpose
PySODEvalToolkit	https://github.com/lartpang/PySODEvalToolkit	Evaluate prediction maps in COD/SOD format
PySODMetrics	https://github.com/lartpang/PySODMetrics	Metric implementation; also installable from PyPI

8.2 Required Metrics

Metric	Direction	Why Required
S-measure (S_m)	Higher better	Structural quality; standard COD metric used by MCOD
E-measure (E_m)	Higher better	Foreground-background alignment; standard COD metric used by MCOD
F-measure (F_m)	Higher better	Precision/recall balance; standard MCOD metric
MAE	Lower better	Pixel-level prediction error; standard MCOD metric
Weighted F-measure (wF)	Higher better	Adds error sensitivity for spatially weighted mistakes; extension metric
Dice	Higher better	Overlap clarity, especially useful for small objects
mIoU	Higher better	General mask overlap metric for foundation-model prompts
Boundary F-score	Higher better	Boundary quality, important for camouflage contours

8.3 Promptability-Specific Metrics

Metric	Definition	Use
Mask IoU	IoU between predicted mask and GT mask	Main FM overlap metric
Prompt Success@0.5	Percentage with mask IoU >= 0.5	How often prompts produce usable masks
Prompt Success@0.7	Percentage with mask IoU >= 0.7	Strict prompt success

Oracle Gap	Oracle GT-box result minus fair generated-prompt result	Quantifies prompt generation bottleneck
Random-Point Sensitivity	Standard deviation across repeated random in-mask prompt trials	Prompt stability diagnostic
Box Prompt Recall	Recall of generated boxes against GT-mask-derived box	Measures prompt localisation quality

8.4 Efficiency Metrics

Metric	Use
Parameters	Model size
FLOPs	Computational complexity where computable
FPS	Throughput
Latency per image	Practical inference cost
GPU peak memory	Reproducibility and feasibility

TASK 9: Establish Reproducible COD Baseline Anchors

9.2 Required Baselines to Run

Priority	Model	Official Repository	Input Conditions	Purpose
1	CamoFormer	HVision-NKU/CamoFormer	Official false-colour 3-channel first; optional 8-channel adaptation after validation	Strong reproducible COD prompt-source model
2	SINet-V2	GewelsJI/SINet-V2	Official false-colour; optional 8-channel adaptation	Reproducibility anchor linked to classic COD
3	ZoomNeXt	lartpang/ZoomNeXt	Official false-colour; optional MSI adaptation	Strong optional modern COD baseline
4	VSCode	Sssssuperior/VSCode	RGB-like view only unless adaptation is	Prompt/generalist related comparison,

			justified	optional

9.3 MSI Adaptation Rule

If adapting an RGB COD baseline to 8-channel MCOD input, document the modification explicitly and follow the original MCOD paper's principle: expand the original 3-channel input layer to eight channels and record the initialisation method.

9.4 Required Baseline Runs

Run ID	Model	Input	Required?	Output Use
CamoFormer_FC	CamoFormer	Official false-colour view	Yes	RGB-like predicted prompt source
CamoFormer_MSI8	CamoFormer modified input	All eight bands	Yes if implementation succeeds	MSI predicted prompt source
SINetV2_FC	SINet-V2	Official false-colour view	Recommended	Validation anchor
SINetV2_MSI8	SINet-V2 modified input	All eight bands	Recommended	Spectral prompt-source comparison
ZoomNeXt_FC	ZoomNeXt	Official false-colour	Optional	Strong RGB comparison

Table 1- Performance Comparison for HSI Data (MCOD Dataset) -

9.5 Baseline Result Table

Model	Input	Trained by Student?	S-measure ↑	E-measure ↑	Weighted F ↑	F-measure ↑	MAE ↓	Dice ↑	mIoU ↑	Use as Prompt Source?
Official MCOD best reported method	MSI	No - reported reference			N/A if unreported			N/A	N/A	No
CamoFormer	False-colour view	Yes	0.7589	0.9025	0.5929	0.6418	0.00353	0.5935	0.4777	RGB-like prompt
CamoFormer	Eight-band MSI	Yes	0.8332	0.9208	0.6905	0.6579	0.00320	0.7124	0.5936	MSI prompt
SINet-V2	False-colour	Yes	0.7893	0.8451	0.5929	0.5554	0.004	0.63	0.509	Optional prompt

	view						25	01	6	
SINet-V2	Eight-band MSI	Yes	0.7893	0.8476	0.5881	0.5522	0.00407	0.6257	0.5066	Optional prompt

Table 2- Performance Comparison for RGB Data (Pseudo-coloured MCOD Dataset) -

9.6 Files to Save

```
outputs/predictions/camoformer_false_colour/

outputs/predictions/camoformer_msi8/

outputs/predictions/sinetv2_false_colour/

outputs/predictions/sinetv2_msi8/

outputs/tables/cod_baseline_anchor_table.csv

outputs/figures/cod_baseline_qualitative_grid.png
```

Table 3- Performance Comparison for RGB Data (Dataset) -

TASK 10: Run the Core Foundation-Model Benchmark - SAM and SAM2

10.1 Models to Run

Model	Official Source	Required?	Input Constraint
SAM ViT-B or ViT-H	facebookresearch/segment-anything	Yes	Three-channel image views
SAM2 Hiera-Small or Base+ for initial screening	facebookresearch/sam2	Yes	Three-channel image views
SAM2 Hiera-Large for final selected experiments	facebookresearch/sam2	If GPU allows	Three-channel image views
SAM2-Adapter	tianrun-chen/SAM-Adapter-PyTorch	If GPU allows	Treat as related trainable RGB-COD baseline; do not call it a new spectral method

10.2 Input Views for Foundation Models

SAM and SAM2 accept image-style three-channel inputs. Therefore, for the zero-shot benchmark, run the following views:

Input View	Construction	Required?	Research Question
Official false-colour	Official S2/S3/S5 composite convention after verification	Yes	How does the normal MCOD view perform?
Red-edge grayscale-3	Normalised S6 repeated to three channels	Yes	Does red-edge alone expose camouflage to FM masks?
NIR1 grayscale-3	Normalised S7 repeated to three channels	Yes	Does NIR1 improve promptability?
NIR2 grayscale-3	Normalised S8 repeated to three channels	Yes	Does NIR2 improve promptability?
NIR group projected view	Documented S6-S8 projection into three channels	Optional	Does joint spectral view help?
All-8 learned projection	A trained 8-to-3 projection; not zero-shot	Strong extension only	Does learned projection improve FM input?

10.3 Prompt Protocols

Run each prompt type separately and label fairness correctly.

Prompt Mode	How to Generate	Uses Ground Truth at Inference?	Fairness Label	Required?
Automatic mask generation	SAM/SAM2 automatic masks, choose prediction under documented matching/selection rule	No GT prompt; selection rule must not use GT for "fair" score	Fair/no-GT only if mask selection is GT-free	Yes
Grid-point automatic protocol	Fixed grid points independent of GT; merge masks with documented rule	No	Fair/no-GT	Recommended
GT centroid point	Point inside GT mask / centroid	Yes	Oracle-lite diagnostic	Yes
Random positive GT point	Random point sampled inside GT mask; repeat K=5	Yes	Oracle-lite sensitivity diagnostic	Yes
GT bounding box	Bounding box derived from GT mask	Yes	Oracle upper bound	Yes
RGB-COD predicted mask to box	Convert false-colour COD model prediction into prompt box	No GT prompt	Fair cascade	Yes

MSI-COD predicted mask to box	Convert 8-band COD model prediction into prompt box	No GT prompt	Fair spectral cascade	Yes if MSI baseline works
Spectral-view consensus prompt	Combine no-GT candidate masks/boxes from S6/S7/S8 by predeclared rule	No GT prompt	Exploratory fair baseline	Optional, must not be marketed as new spectral-SAM method without novelty review

10.6 Core Foundation-Model Result Table

Model	Input View	Prompt Type	Fair/Oracle	S-measure ↑	E-measure ↑	wF ↑	MAE ↓	Dice ↑	mIoU ↑	Success@0.5 ↑	Oracle Gap ↓
SAM	False-colour	Automatic/grid	Fair								
SAM2	False-colour	Automatic/grid	Fair								
SAM2	False-colour	GT box	Oracle								
SAM2	NIR1	GT box	Oracle								
SAM2	False-colour	RGB-COD predicted box	Fair cascade								
SAM2	False-colour	MSI-COD predicted box	Fair spectral cascade								

10.7 Output Files

```
outputs/predictions/foundation_models/SAM/

outputs/predictions/foundation_models/SAM2/

outputs/prompts/prompt_records.csv

outputs/metrics/foundation_model_promptability_results.csv

outputs/tables/table_fm_promptability.csv

outputs/figures/fm_promptability_examples.png
```

TASK 11: Run Spectral View and Band-Group Analysis

11.1 Goal

Measure whether red-edge and NIR views reveal information that affects foundation-model segmentation behaviour,

11.2 Required View Analysis

Experiment ID	View	Prompt Protocol	Required?
V1	Official false-colour	GT box oracle and fair generated box	Yes
V2	S6 red-edge repeated	GT box oracle and fair generated box	Yes
V3	S7 NIR1 repeated	GT box oracle and fair generated box	Yes
V4	S8 NIR2 repeated	GT box oracle and fair generated box	Yes
V5	S6-S8 projected composite	Same protocol	Optional
V6	S1-S5 visible projection	Same protocol	Optional

11.3 Result Table

Model	View	Bands Used	Prompt Type	S-measure ↑	E-measure ↑	wF ↑	MAE ↓	Dice ↑	mIoU ↑	Best Attributes
SAM2	Official false-colour	S2/S3/S5	GT box							
SAM2	Red-edge	S6	GT box							
SAM2	NIR1	S7	GT box							
SAM2	NIR2	S8	GT box							
SAM2	Official false-colour	S2/S3/S5	MSI-COD predicted box							

11.4 Attribute-Wise Spectral View Table

View / Prompt	SO ↑	BS ↑	CB ↑	OE ↑	II ↑	UI ↑	OC ↑	SC ↑	Overall S-measure ↑
False-colour									

+ GT box									
Red-edge + GT box									
NIR1 + GT box									
NIR2 + GT box									
False-colour + MSI predicted box									

TASK 12: Create the Spectral-Degradation Robustness Benchmark

12.1 Purpose

This is one of the central new benchmark contributions. Foundation models may behave very differently when spectral evidence is incomplete, noisy or spatially misaligned.

12.3 Corruptions to Test

Corruption Group	Corruption	Severity 1	Severity 2	Severity 3	Scientific Purpose
Visible degradation	False-colour low illumination	Brightness x 0.7	x 0.5	x 0.3	Test poor-visible-light failure
Visible degradation	Overexposure	Brightness x 1.2	x 1.5	x 1.8	Test loss of texture/contours
Sensor degradation	Gaussian noise per band	sigma 0.01	0.03	0.05 after normalisation	Test spectral sensor noise
Optical degradation	Motion blur on all views	Kernel 5	11	21	Test motion-induced cue loss
Optical degradation	Defocus blur on all views	Kernel 5	9	15	Test boundary loss
Compression	JPEG on three-channel FM views	Quality 70	40	20	Test stored/transmitted images
Spectral missingness	Drop red-edge S6	Replace with zero or training mean	-	-	Determine red-edge dependence
Spectral missingness	Drop NIR1 S7	Replace with zero or training mean	-	-	Determine NIR1 dependence

Spectral missingness	Drop NIR2 S8	Replace with zero or training mean	-	-	Determine NIR2 dependence
Spectral missingness	Drop S6-S8 group	Replace all with training means	-	-	Simulate loss of spectral evidence
Spectral misregistration	Shift S6/S7/S8 spatially	2 px	5 px	10 px	Test registration sensitivity
Spectral scaling	Gain perturbation on S6-S8	+/-5%	+/-10%	+/-20%	Test radiometric sensitivity

12.4 Separate Robustness Protocols

Protocol	Meaning	Models Evaluated
FM-view robustness	Corrupt the three-channel view consumed by SAM/SAM2	SAM, SAM2
Prompt-source robustness	Corrupt input to COD model generating predicted prompt box	CamoFormer/SINet-V2 -> SAM2 cascade
MSI model robustness	Apply band missingness/misalignment to 8-channel COD prompt-source model	MSI CamoFormer/SINet-V2 adaptation
Oracle input sensitivity	Use GT box prompt but corrupt the view	SAM2 only; isolates segmentation sensitivity from prompt localisation

12.5 Metrics

For high-is-better metrics:

```
Relative Drop (%) = 100 * (Clean Score - Corrupted Score) / Clean Score
```

For MAE:

```
Relative MAE Increase (%) = 100 * (Corrupted MAE - Clean MAE) / Clean MAE
```

12.7 Main Robustness Table

Model / Pipeline	Input/Prompt	Clean S ↑	Low-Light S ↑	Noise S ↑	Blur S ↑	JPEG S ↑	Missing NIR S ↑	Misalignment S ↑	Mean Relative Drop ↓
SAM2	False-colour +						N/A	N/A	

	GT box oracle								
SAM2	False-colour + RGB-COD box						N/A	N/A	
SAM2	False-colour + MSI-COD box								
MSI COD prompt-source model	8-band mask prediction					N/A			

12.8 Band-Drop Table

Pipeline	Clean	Drop S6 Red-Edge	Drop S7 NIR1	Drop S8 NIR2	Drop S6-S8 Group	Most Critical Band/Group
MSI COD baseline						
MSI-COD -> SAM2 prompt cascade						

12.9 Misalignment Table

Pipeline	Alignment Condition	S-measure ↑	E-measure ↑	wF ↑	MAE ↓	Dice ↑	Relative Drop ↓
MSI COD baseline	Clean aligned						
MSI COD baseline	Shift 2 px						
MSI COD baseline	Shift 5 px						
MSI COD baseline	Shift 10 px						

TASK 13: Attribute-Wise and Failure Analysis

13.1 Required Attribute Analysis

Use the eight official MCOD challenge attributes. For each foundation-model pipeline, report performance on each attribute subset.

Pipeline	SO	BS	CB	OE	II	UI	OC	SC	Overall
SAM2 false-colour automatic/grid									
SAM2 false-colour GT box oracle									
SAM2 RGB-COD predicted box									
SAM2 MSI-COD predicted box									

13.2 Failure Taxonomy

Failure Type	What It Means	How to Extract Examples
Very small target missed	GT mask area ratio is extremely small; prompt/mask misses object	Rank SO subset by lowest IoU
Boundary invisible in false-colour	RGB-like view misses contour but spectral view helps	Compare false-colour and NIR prompt outputs
Spectral cue not helpful	Object and background remain similar in red-edge/NIR	Find cases failing across all views
Illumination failure	OE, II or UI subset causes mask collapse	Rank extreme-illumination drop
Automatic-mask selection failure	SAM/SAM2 produces candidate but GT-free selector chooses wrong mask	Compare oracle best candidate versus fair selected candidate
Predicted-box prompt failure	COD source mask gives poor box, leading SAM2 to fail	Box recall and mask IoU analysis
SAM2 under-segmentation	Mask finds only a fragment	Rank high precision / low recall masks
SAM2 over-segmentation	Mask leaks into similar background	Rank large false-positive region masks
Missing-band failure	Pipeline collapses when NIR/red-edge removed	Rank drop under band ablation
Misalignment failure	Spatially shifted spectral band causes wrong prompt/mask	Rank drop under 10-pixel shift

TASK 14: Efficiency and Reproducibility Analysis

14.1 Efficiency Measurements

Measure under one declared hardware and software environment:

Measurement	Requirement
GPU model and memory	Record exactly
CPU and RAM	Record exactly
PyTorch and CUDA version	Record exactly
Input resolution	Declare for each method
Batch size for FPS	Use batch size 1 and optionally batch size 4 separately
Warm-up	At least 20 iterations
Timed iterations	At least 100 images or all test images
Parameters	Measure for trainable/model parameters
Peak GPU memory	Measure inference peak

14.2 Efficiency Table

Model / Pipeline	Input View	Prompt Type	Params M ↓	FLOPs G ↓	FPS ↑	Latency ms/image ↓	Peak GPU Memory GB ↓	S-measure ↑	MAE ↓
SAM	False-colour	GT box oracle							
SAM2	False-colour	GT box oracle							
SAM2	False-colour	RGB-COD box							
SAM2	False-colour	MSI-COD box							

14.3 Seed and Statistical Reporting

Zero-shot SAM/SAM2 results are deterministic only if prompt generation and selection are deterministic. Any stochastic process must use fixed seeds.

For trained COD prompt-source models, run at least:

```
Seeds: 42, 123, 3407
```

Report:

```
mean +/- standard deviation
```

For the primary comparison between RGB-COD prompts and MSI-COD prompts, compute a paired bootstrap confidence interval over test images for mIoU or S-measure improvement.
Will look after this for further extension, depends on the results.

INCLUDE THE RESULTING IMAGES, DATASET IMAGES

10 Recent, Important, and Credible Papers on Camouflage Pattern / Image Generation

Prepared for background-conditioned camouflage generation research | Verified online: 12 June 2026

Scope: This list prioritizes papers useful for generating background-adaptive camouflage textures/images, synthetic camouflaged datasets, and controllable camouflage generation pipelines. Preprints are marked clearly; peer-reviewed journal/conference papers are prioritized where available.

Best first-reading order for your project: LAKE-RED -> CamoX -> CT-CIG -> Foreground Focus/FACIG -> RealCamo -> GenCAMO.

A. Quick List

Most recent diffusion / generation papers (1-5)

Rank	Year	Paper	Venue / Status	Why important	Link
1	2026	CamoX: A Diffusion-Based Method With Few-Shot Learning for Environment-Guided Camouflage Pattern Generation	IEEE Transactions on Artificial Intelligence, 7(4):1956-1971, Apr. 2026. DOI: 10.1109/TAI.2025.3608758	Best direct match: diffusion + few-shot environment adaptation; introduces Camo-Meta dataset.	Open
2	2026	GenCAMO: Scene-Graph Contextual Decoupling for Environment-aware and Mask-free Camouflage Image-Dense Annotation Generation	arXiv preprint, Jan. 2026	Environment-aware synthetic camouflage data with dense annotations, depth, scene graphs, and prompts.	Open

3	2026	AI-Driven Adaptive Camouflage Pattern Generation for Helicopter Detection Evasion in Aerial Sensor Imagery Using Fine-Tuned YOLOv8 and Stable Diffusion	Sensors, 26(6), Article 1895, Mar. 2026. DOI: 10.3390/s26061895	Object-adaptive aircraft/helicopter camouflage pipeline using YOLO masks + K-means + Stable Diffusion.	Open
4	2025	RealCamo: Boosting Real Camouflage Synthesis with Layout Controls and Textual-Visual Guidance	arXiv preprint, Dec. 2025	Adds layout controls and text-visual guidance for realistic camouflage synthesis.	Open
5	2025 / 2026	Text-guided Controllable Diffusion for Realistic Camouflage Images Generation	arXiv preprint, Nov. 2025; associated code lists AAAI 2026	Text-guided controllable diffusion with VLM prompts and object location/shape control.	Open

A. Quick List continued

Additional important recent / baseline papers (6-10)

Rank	Year	Paper	Venue / Status	Why important	Link
6	2025	Foreground Focus: Enhancing Coherence and Fidelity in Camouflaged Image Generation	arXiv preprint, Apr. 2025	Foreground-aware generation that improves object fidelity and foreground-background coherence.	Open
7	2025	Digital Camouflage Pattern Generation Approach Based on Fragmentation Edges	2025 6th International Conference on Electrical, Electronic Information and Communication Engineering (EEICE). DOI: 10.1109/EEICE65049.2025.11034215	Recent digital pattern method; useful non-diffusion baseline for edge/fragmentation-based designs.	Open
8	2024	LAKE-RED: Camouflaged Images Generation by Latent Background Knowledge Retrieval-Augmented Diffusion	CVPR 2024; arXiv 2404.00292	Core CVPR paper: retrieval-augmented diffusion and latent background knowledge for CIG.	Open
9	2023 / 2024	CamDiff: Camouflage Image Augmentation via Diffusion Model	arXiv 2304.05469; published version indexed by Artificial Intelligence and Robotics / SciOpen	Early latent-diffusion camouflage augmentation method using CLIP to reduce prompt failures.	Open
10	2023	Adaptive Camouflage Pattern Generation to Different Environments Via Content-Aware Style Transfer	IEEE International Conference on Image Processing (ICIP) 2023. DOI: 10.1109/ICIP49359.2023.10221991	Environment-adaptive camouflage pattern generation via content-aware style transfer.	Open

B. Notes for Each Paper

1. CamoX: A Diffusion-Based Method With Few-Shot Learning for Environment-Guided Camouflage Pattern Generation

Year: 2026 | **Status:** Peer-reviewed journal

Authors: Tran Thanh Phong Nguyen, Tauseef Gulrez, Joanne B. Culpepper, Son Lam Phung, Thanh Le Hoang

Venue / DOI: IEEE Transactions on Artificial Intelligence, 7(4):1956-1971, Apr. 2026. DOI: 10.1109/TAI.2025.3608758

Why it matters: Strongest direct match for environment-guided camouflage pattern generation. Uses latent diffusion plus few-shot adaptation so generated patterns can adapt to new environments with limited retraining. Introduces the Camo-Meta dataset of 144,750 realistic camouflage patterns and metadata.

How to use it in your project: Best implementation starting point if your final goal is background-conditioned camouflage texture generation.

Link: https://doi.org/10.1109/TAI.2025.3608758

2. GenCAMO: Scene-Graph Contextual Decoupling for Environment-aware and Mask-free Camouflage Image-Dense Annotation Generation

Year: 2026 | **Status:** Recent preprint

Authors: Chenglizhao Chen, Shaojiang Yuan, Xiaoxue Lu, Mengke Song, Jia Song, Zhenyu Wu, Wenfeng Song, Shuai Li

Venue / DOI: arXiv preprint, Jan. 2026

Why it matters: Very recent environment-aware camouflage generation framework. Important because it connects camouflage synthesis with dense annotations, depth maps, scene graphs, attribute descriptions, and text prompts.

How to use it in your project: Useful if you want synthetic datasets with masks/annotations for COD or MCOD benchmarking.

Link: https://arxiv.org/abs/2601.01181

3. AI-Driven Adaptive Camouflage Pattern Generation for Helicopter Detection Evasion in Aerial Sensor Imagery Using Fine-Tuned YOLOv8 and Stable Diffusion

Year: 2026 | **Status:** Peer-reviewed journal

Authors: Jonghyeok Im, Yeonhong Kim, Heoung-Jae Chun, Kyoungsik Kim

Venue / DOI: Sensors, 26(6), Article 1895, Mar. 2026. DOI: 10.3390/s26061895

Why it matters: Directly military-relevant because it studies adaptive camouflage generation for helicopters in aerial sensor imagery. Combines YOLO-based masking, K-means background color extraction, and Stable Diffusion inpainting.

How to use it in your project: Useful for understanding object-specific camouflage generation pipelines, especially for aircraft/vehicle-like objects.

Link: https://doi.org/10.3390/s26061895

4. RealCamo: Boosting Real Camouflage Synthesis with Layout Controls and Textual-Visual Guidance

Year: 2025 | **Status:** Recent preprint

Authors: Chunyuan Chen, Yunuo Cai, Shujuan Li, Weiyun Liang, Bin Wang, Jing Xu

Venue / DOI: arXiv preprint, Dec. 2025

Why it matters: Recent CIG paper focused on realistic camouflage synthesis. Adds layout control and textual-visual guidance to improve semantic coherence between foreground objects and generated backgrounds.

How to use it in your project: Good for your problem because you need object placement, object shape, and background-consistent texture control.

Link: https://arxiv.org/abs/2512.22974

5. Text-guided Controllable Diffusion for Realistic Camouflage Images Generation

Year: 2025 / 2026 | **Status:** Recent preprint / conference-associated code

Authors: Yuhang Qian, Haiyan Chen, Wentong Li, Ningzhong Liu, Jie Qin

Venue / DOI: arXiv preprint, Nov. 2025; associated code lists AAAI 2026

Why it matters: Introduces CT-CIG, a controllable text-guided camouflage image generation method. Uses visual-language models for text prompt construction, Stable Diffusion fine-tuning, and a lightweight controller for object location/shape.

How to use it in your project: Very useful if you want inputs like background + object class + text prompt + object location.

Link: https://arxiv.org/abs/2511.20218

6. Foreground Focus: Enhancing Coherence and Fidelity in Camouflaged Image Generation

Year: 2025 | **Status:** Recent preprint

Authors: Pei-Chi Chen, Yi Yao, Chan-Feng Hsu, HongXia Xie, Hung-Jen Chen, Hong-Han Shuai, Wen-Huang Cheng

Venue / DOI: arXiv preprint, Apr. 2025

Why it matters: Proposes a foreground-aware CIG model that improves foreground-background coherence and preserves small/foreground object fidelity. It directly addresses color discrepancy and foreground distortion problems in camouflage generation.

How to use it in your project: Important when your object is a soldier, vehicle, tank, or aircraft and must not become distorted during camouflage synthesis.

Link: https://arxiv.org/abs/2504.02180

7. Digital Camouflage Pattern Generation Approach Based on Fragmentation Edges

Year: 2025 | **Status:** Peer-reviewed conference paper

Authors: Liqiu Li, Mingjun Li, Congwei Liao, Lianwen Deng

Venue / DOI: 2025 6th International Conference on Electrical, Electronic Information and Communication Engineering (EEICE). DOI: 10.1109/EEICE65049.2025.11034215

Why it matters: Recent digital camouflage pattern generation paper. Useful as a classical/computer-vision-style complement to diffusion methods because it focuses on fragmentation edges and digital pattern design.

How to use it in your project: Use this to understand pixel/digital camouflage structure, edge fragmentation, and non-diffusion baselines.

Link: https://doi.org/10.1109/EEICE65049.2025.11034215

8. LAKE-RED: Camouflaged Images Generation by Latent Background Knowledge Retrieval-Augmented Diffusion

Year: 2024 | **Status:** Top-tier CVPR paper

Authors: Pancheng Zhao, Peng Xu, Pengda Qin, Deng-Ping Fan, Zhicheng Zhang, Guoli Jia, Bowen Zhou, Jufeng Yang

Venue / DOI: CVPR 2024; arXiv 2404.00292

Why it matters: Probably the most important foundation paper for your project. It uses retrieval-augmented diffusion and background knowledge reasoning for camouflaged image generation.

How to use it in your project: Study this first for the main idea of using background knowledge to guide camouflage generation.

Link: https://arxiv.org/abs/2404.00292

9. CamDiff: Camouflage Image Augmentation via Diffusion Model

Year: 2023 / 2024 | **Status:** Diffusion-based augmentation paper

Authors: Xiaojun Luo et al.

Venue / DOI: arXiv 2304.05469; published version indexed by Artificial Intelligence and Robotics / SciOpen

Why it matters: Early diffusion-based camouflage generation/augmentation work. Uses latent diffusion to synthesize objects in camouflaged scenes and CLIP to reduce prompt-synthesis failures.

How to use it in your project: Useful for understanding how synthetic camouflage images can improve training data diversity for COD models.

Link: https://arxiv.org/abs/2304.05469

10. Adaptive Camouflage Pattern Generation to Different Environments Via Content-Aware Style Transfer

Year: 2023 | **Status:** Peer-reviewed IEEE conference paper

Authors: Min-jae Kim, Subin Kwon, Byung-hyun Ahn, Euntaek Ha, Yeonhee Choi, Joonki Paik

Venue / DOI: IEEE International Conference on Image Processing (ICIP) 2023. DOI: 10.1109/ICIP49359.2023.10221991

Why it matters: Direct environment-adaptive camouflage pattern generation method using content-aware style transfer. Useful bridge between classical style-transfer camouflage and newer diffusion methods.

How to use it in your project: Good baseline idea if you want a lighter, non-diffusion system for adapting patterns to terrain/background images.

Link: https://doi.org/10.1109/ICIP49359.2023.10221991

C. Recommended Grouping for Implementation

- **Core diffusion / background-conditioning papers:** LAKE-RED, CamoX, CT-CIG, RealCamo, Foreground Focus/FACIG
- **Synthetic dataset generation papers:** GenCAMO, CamDiff, LAKE-RED
- **Military/object-specific camouflage generation:** AI-Driven Adaptive Camouflage Pattern Generation for Helicopter Detection Evasion; Digital Fragmentation Edges paper
- **Classical / lightweight baselines:** Adaptive Content-Aware Style Transfer; Digital Fragmentation Edges paper

D. Important Note

Use boundary: This list is meant for academic research, synthetic data generation, benchmarking, and understanding camouflage generation methods. For implementation, keep experiments in a controlled research setting and evaluate with standard image-quality and camouflage-detection metrics rather than real-world deployment claims.

Project Titel

AI-based Camouflage Image creation and Detection for Images

Dataset Used

MCOD Dataset

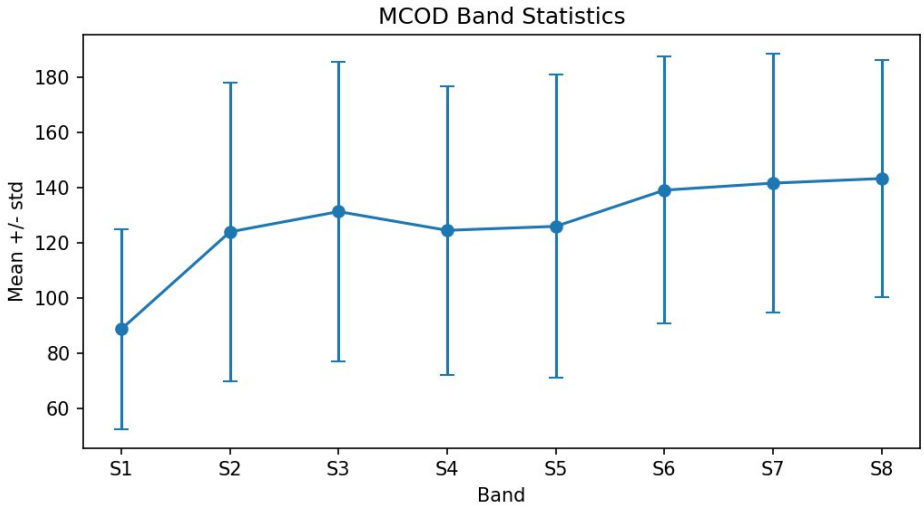
Official link: https://github.com/yl2900260-bit/MCOD

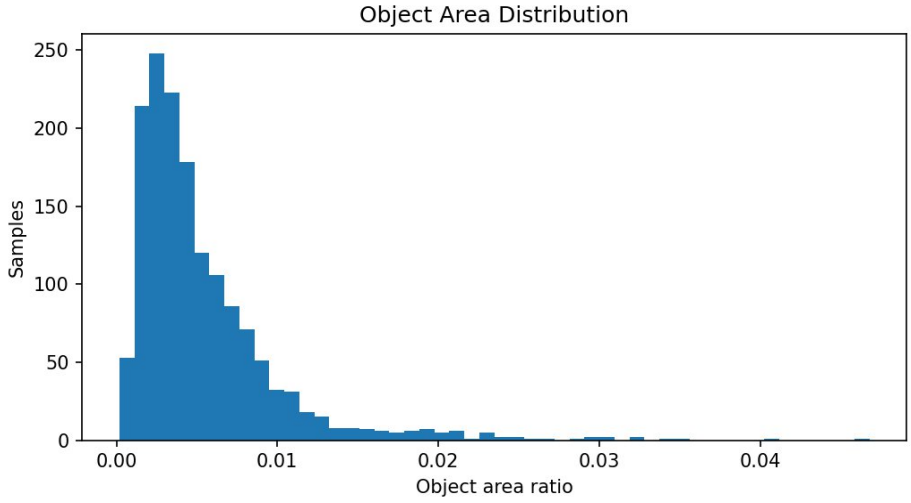
Total samples	The dataset contains **1,527 multispectral camouflaged object samples**.
Training samples	**1,027 samples** are used for training the detection/generation models.
Testing samples	**500 samples** are used for testing and evaluating model performance.
Input data	Each sample includes **official false-colour RGB images** and corresponding **8-band multispectral images**.

Annotations	Each image has a **pixel-level ground-truth mask**, which marks the exact camouflaged object region.
False-colour RGB use	The false-colour images are used as the first-stage input because most existing COD and generation models are designed for 3-channel RGB images.
8-band multispectral use	The 8-band images provide additional spectral information beyond RGB, which can help study whether the object remains detectable in hidden spectral bands.
Main challenge	Many objects in MCOD are **small, weak-boundary, and highly similar to the background**, making both detection and generation difficult.
Use in detection	The dataset is used to train and evaluate COD models such as **SINet-V2, CamoFormer, and ZoomNet** on false-colour RGB and later 8-band MSI input.
Use in generation	The masks and images are used to guide camouflage generation/inpainting models so that the object region can be modified while preserving the background.

Relevance to this project	MCOD is suitable for studying **small-object-aware camouflage generation**, **false-colour RGB baseline generation**, and later **multispectral-aware camouflage refinement**.







Brief Literature Review

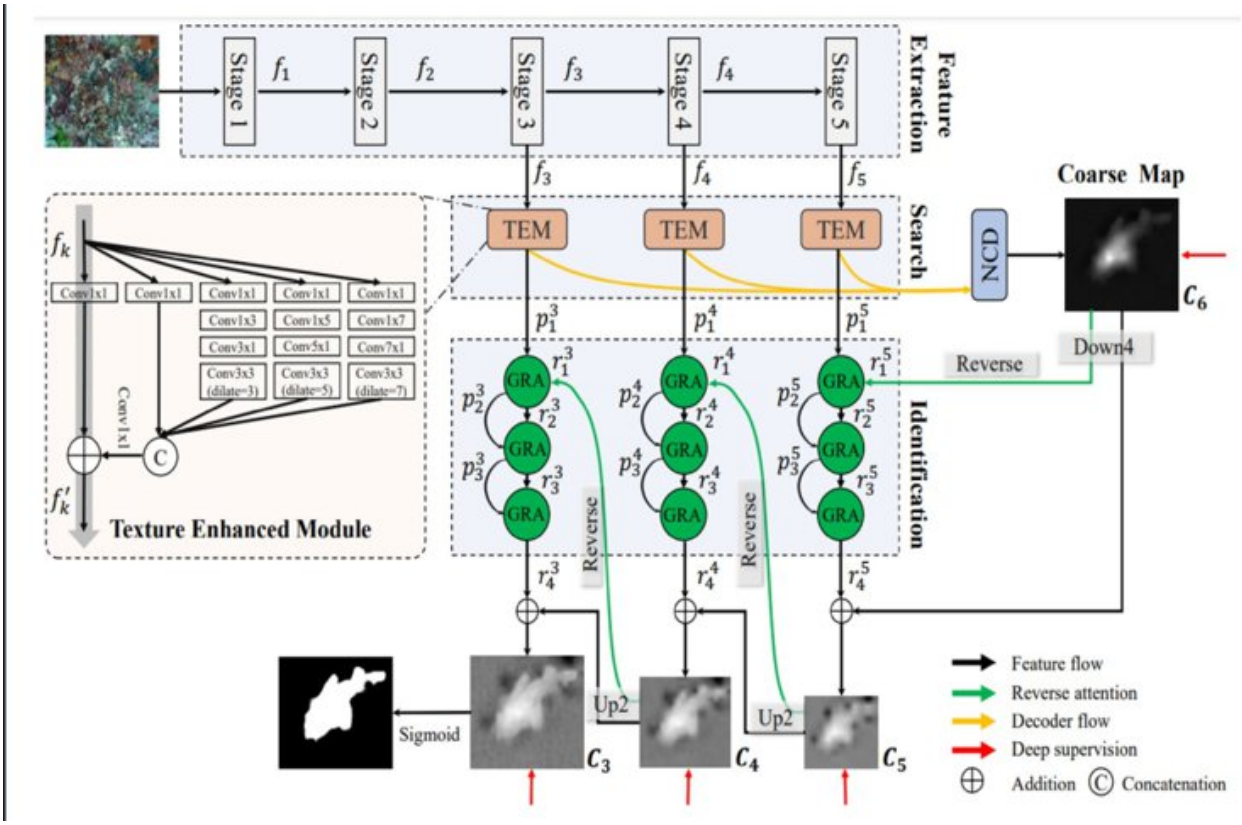
Image detection:

The official MCOD work introduced the first dedicated eight-band multispectral benchmark for camouflaged object detection. It evaluated eleven representative COD methods and reported that several methods benefited from multispectral input.

Models used:

Brief Literature Review: Detection and Generation Models

1. Fan et al. (2022) — SINet-V2: Search and Identification Network Version 2



Review:
SINet-V2 is an improved version of the original SINet model for camouflaged object detection. It was proposed for concealed object detection, where objects are visually blended with the background due to similar colour, texture, shape, and boundary appearance. The model follows a biologically inspired search-and-identification strategy. In the search stage, it locates possible camouflaged object regions, and in the identification stage, it refines these regions to produce the final segmentation mask.

Methods Used:
SINet-V2 uses a Res2Net-based convolutional backbone pretrained on ImageNet. Two important modules are introduced: the Neighbor Connection Decoder and Group-Reversal Attention. The Neighbor Connection Decoder combines neighbouring high-level features to generate a coarse localization map. Group-Reversal Attention divides feature channels into groups and uses reverse attention to

progressively refine uncertain regions and object boundaries. The model was evaluated on standard RGB COD datasets such as CAMO, COD10K, CHAMELEON, and NC4K.

Advantages:

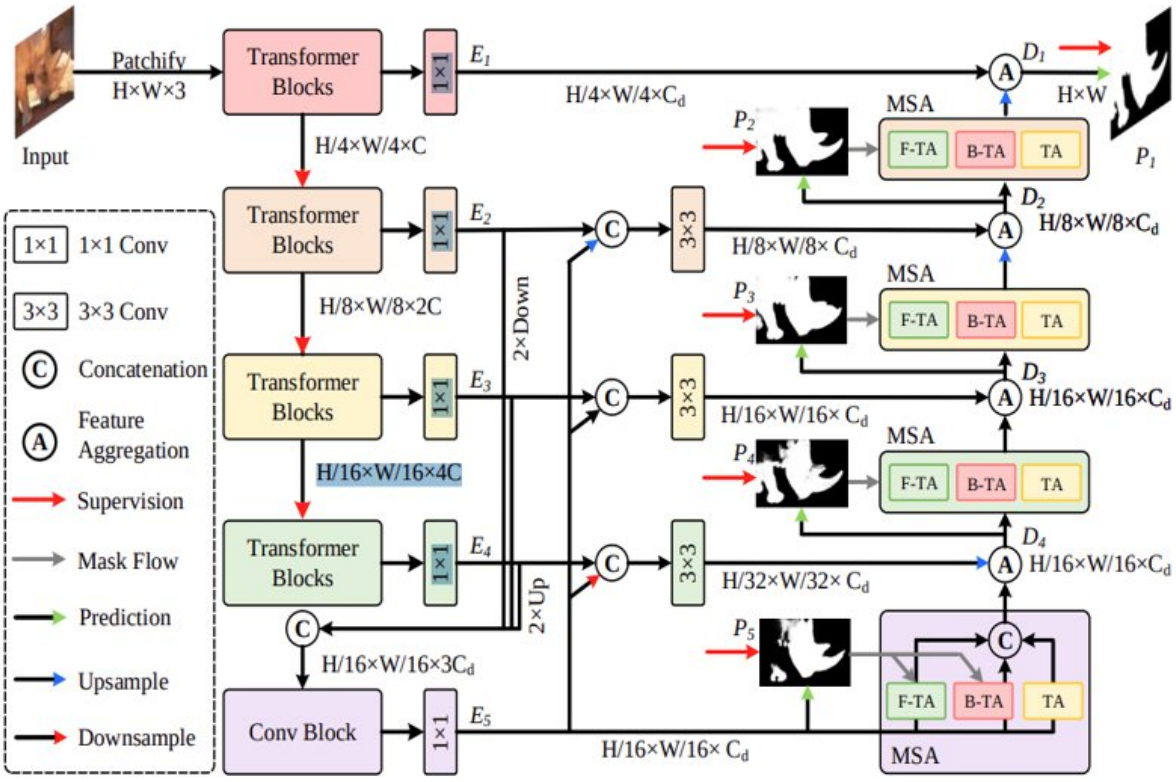
- Strong RGB camouflaged object detection baseline.
- Improves over the original SINet using better feature decoding and reverse attention refinement.
- Useful for detecting objects with weak boundaries and high background similarity.
- Official implementation and pretrained weights are available, making it practical for baseline comparison.
- Can be adapted from 3-channel false-colour RGB input to 8-band multispectral input by modifying the first input layer.

Limitations:

- Originally designed for RGB images, not multispectral imagery.
- Uses ImageNet/RGB-based pretraining, which may not fully transfer to 8-band MCOD data.
- Small camouflaged objects may still be missed because the model mainly relies on high-level semantic features.
- Boundary localization can become weak when the object is tiny or strongly blended with the background.
- Does not explicitly use spectral-band relationships or band-wise fusion, which are important for MCOD.

Relevance to Present Work:
SINet-V2 is useful as a strong RGB COD baseline for the official false-colour MCOD images. It can also be modified for 8-band input to compare whether full multispectral information improves detection over false-colour RGB.

2. Yin et al. (2024) — CamoFormer: Masked Separable Attention for Camouflaged Object Detection



Review:
CamoFormer is a transformer-based camouflaged object detection model that addresses the difficulty of separating camouflaged objects from highly similar backgrounds. Unlike CNN-based methods, it uses transformer attention to capture long-range dependencies and global context. The key motivation is that normal self-attention may treat foreground and background regions similarly, which is problematic in camouflage scenes.

Methods Used:
CamoFormer introduces Masked Separable Attention, which separates the modeling of camouflaged object features and background features using different mask strategies. It uses a backbone encoder, MSA-based feature modeling, and a lightweight top-down decoder to progressively recover high-resolution semantic representations and generate the final segmentation map. The model was evaluated on widely used RGB COD datasets including CAMO, COD10K, and CHAMELEON.

Advantages:

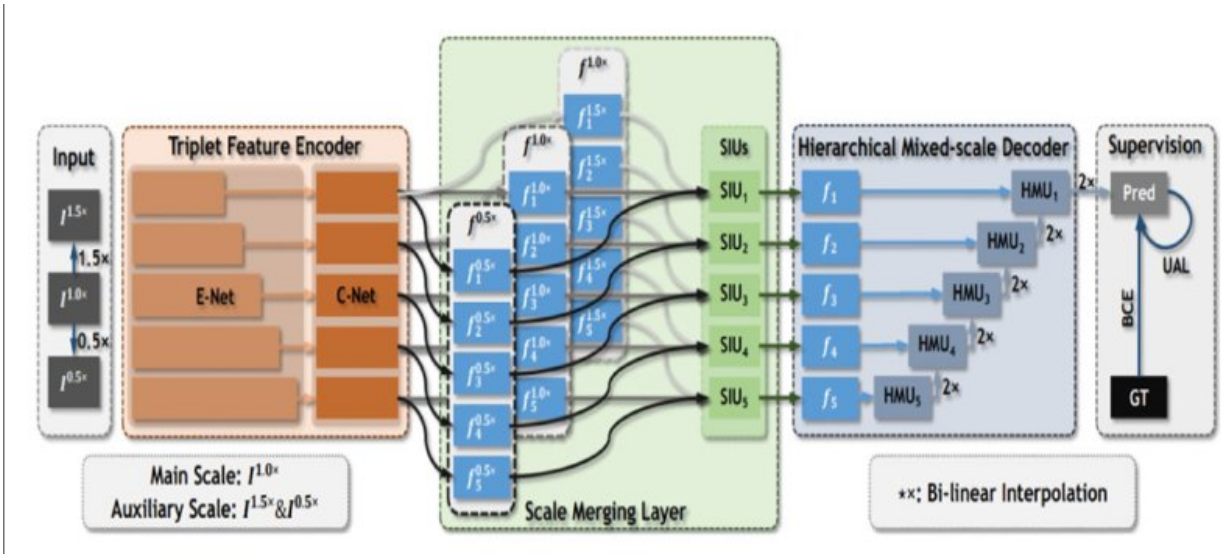
- Transformer-based architecture captures long-range contextual relationships better than many CNN-based models.
- Masked Separable Attention explicitly separates foreground and background feature modeling.
- Achieves strong performance on standard COD benchmarks.
- Better suited for complex backgrounds where local CNN features may be insufficient.
- Can be used as a strong modern baseline for MCOD false-colour RGB detection.

Limitations:

- Originally designed for RGB camouflaged object detection, not 8-band multispectral data.
- Transformer models may require more memory and training data compared with simpler CNN-based models.
- Small-object detection may still be challenging because tiny objects can be suppressed during patch/token-based feature processing.
- Does not explicitly model spectral signatures across multiple bands.
- Input embedding must be modified carefully for 8-band MCOD adaptation.

Relevance to Present Work:
CamoFormer is relevant because it provides a strong attention-based COD baseline. In this project, it can be trained on false-colour MCOD images and later adapted to 8-band input by modifying the input embedding layer. Its foreground-background attention design is useful for studying whether transformer attention improves camouflaged object detection in multispectral scenes.

3. Pang et al. (2022) — ZoomNet: Zoom In and Out: A Mixed-scale Triplet Network for Camouflaged Object Detection



Review:
ZoomNet is a mixed-scale camouflaged object detection model inspired by the human visual process of zooming in and zooming out while observing unclear objects. Camouflaged objects often appear at different scales, have fuzzy boundaries, and may be partially occluded. A single fixed-scale representation may fail to capture both global context and fine object details. ZoomNet addresses this problem using mixed-scale semantic learning.

Methods Used:
ZoomNet uses a mixed-scale triplet network that extracts and combines features at different visual scales. Large-scale features help understand global object location and scene context, while small-scale features help recover fine object details and weak boundaries. It introduces scale integration and hierarchical mixed-scale processing to combine multi-level information. The method also uses an uncertainty-aware loss to reduce ambiguous predictions in difficult regions.

Advantages:

- Strong baseline for objects appearing at different sizes.
- Useful for datasets like MCOD where object size varies and many targets are small.
- Zoom-in/zoom-out strategy helps combine global context with local boundary information.
- Uncertainty-aware loss helps handle ambiguous camouflaged regions.
- Official code is available, making it practical for comparison.

Limitations:

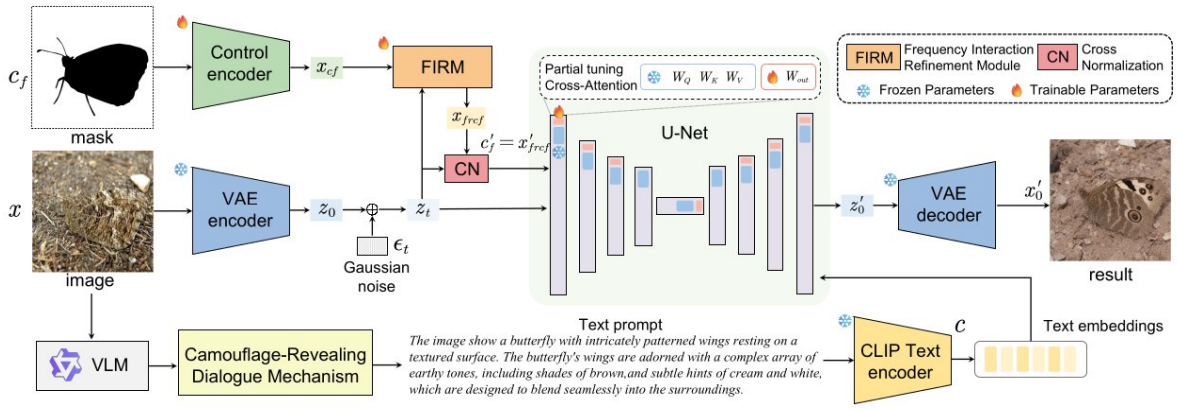
- Designed for RGB COD datasets and does not naturally exploit multispectral bands.
- Multi-scale processing can increase computational cost.
- Very small objects may still lose detail after resizing or downsampling.
- Does not include a dedicated spectral-fusion module for 8-band images.
- Performance may depend strongly on training resolution and crop strategy.

Relevance to Present Work:
ZoomNet is highly relevant because MCOD contains many small and weak-boundary objects. Its mixed-scale design makes it useful for testing whether scale-aware RGB COD models generalize to false-colour MCOD images. Later, it can be modified for 8-band input to study whether multispectral information improves mixed-scale camouflaged object detection.

Image creation/generation

Camouflage image generation methods aim to create foreground objects or textures that match the colour, texture and structure of a background. Earlier methods used colour clustering, procedural textures and GAN-based image translation. Recent diffusion-based approaches use background images, masks and text prompts to generate more realistic and controllable camouflage images

4. Qian et al. (2026) — CT-CIG: Text-guided Controllable Diffusion for Realistic Camouflage Images Generation



Review:
CT-CIG is a camouflage image generation framework rather than a detection model. Its goal is to synthesize realistic camouflaged images where the target object is visually consistent with the

surrounding environment. It addresses the problem that earlier camouflage generation methods may create visually unnatural results because they do not sufficiently model the logical relationship between the object and the background.

Methods Used:
CT-CIG is based on a diffusion generation framework. It uses text prompts, object masks, and diffusion-based control to guide camouflage image synthesis. A Camouflage-Revealing Dialogue Mechanism is used with a vision-language model to generate high-quality prompts for camouflaged images. The method also uses a lightweight controller to guide object location and shape, and a Frequency Interaction Refinement Module to improve high-frequency camouflage textures and local details.

Advantages:

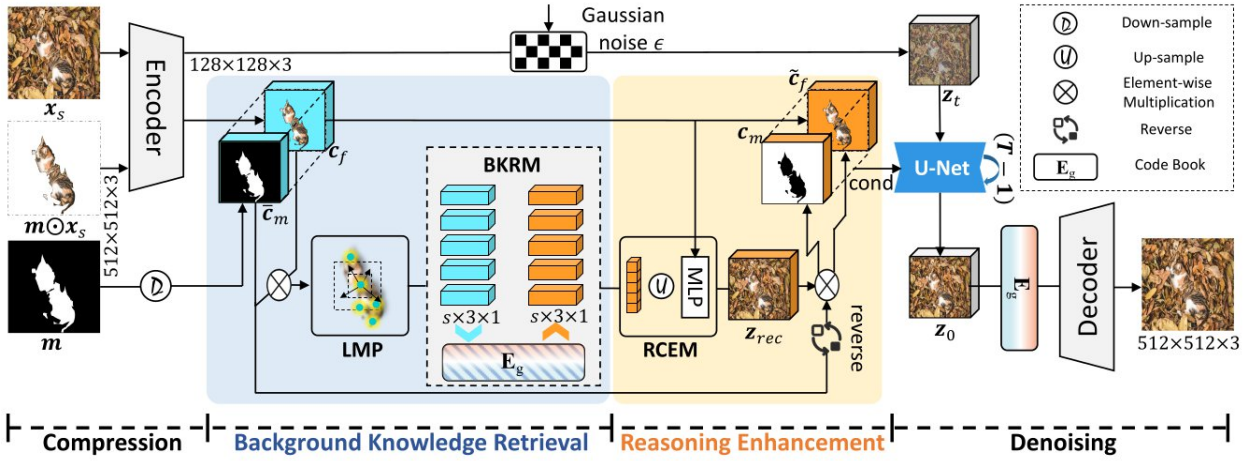
- Specifically designed for camouflage image generation.
- Uses text prompts and masks to provide controllable generation.
- Produces visually realistic and logically plausible camouflage images.
- Frequency refinement helps improve texture-level camouflage details.
- Useful for synthetic data generation and stress-testing COD models.

Limitations:

- Generation quality depends strongly on prompt quality and mask guidance.
- Target shape and boundary accuracy may not always be reliable for small objects.
- Mainly focuses on RGB-style visual camouflage and does not directly solve 8-band multispectral camouflage generation.
- Does not explicitly guarantee spectral consistency between object and background in hidden bands.
- For MCOD, small-object-heavy data may require crop-based generation and mask refinement before CT-CIG can work effectively.

Relevance to Present Work:
CT-CIG is relevant because the project focuses on camouflage image generation and synthetic-data creation. However, for MCOD, CT-CIG should be treated as a generation baseline rather than the final solution. Since MCOD has many small objects and 8-band multispectral data, the method needs additional small-object-aware cropping, stronger mask control, and multispectral spectral matching to generate more useful camouflage samples.

5. **Zhao et al. (2024) — LAKE-RED: Camouflaged Images Generation by Latent Background Knowledge Retrieval-Augmented Diffusion**



Review:

LAKE-RED is a diffusion-based camouflage image generation model. Its main goal is to generate realistic camouflaged images by hiding a foreground object inside a visually compatible background. Unlike simple image blending or direct inpainting methods, LAKE-RED tries to retrieve useful background knowledge in the latent space so that the generated camouflage looks more natural and scene-consistent.

Methods Used:

LAKE-RED uses a latent diffusion inpainting framework. It takes a foreground object image and its mask as input, then generates a camouflaged image by completing the surrounding/background region. The key idea is **Latent Background Knowledge Retrieval-Augmented Diffusion**, where background-related latent information is retrieved and used to guide the diffusion generation process. This helps the model produce backgrounds that are semantically and visually compatible with the target object.

Advantages:

- Specifically designed for camouflaged image generation.
- Uses diffusion-based generation, which improves realism compared with traditional synthesis methods.
- Does not require manually designed camouflage patterns.
- Can generate visually natural object-background combinations. Useful as a strong baseline for synthetic camouflage data generation.
- Official code and pretrained checkpoints are available, making reproduction feasible.
- Uses datasets related to COD benchmarks such as CAMO, COD10K, and NC4K.

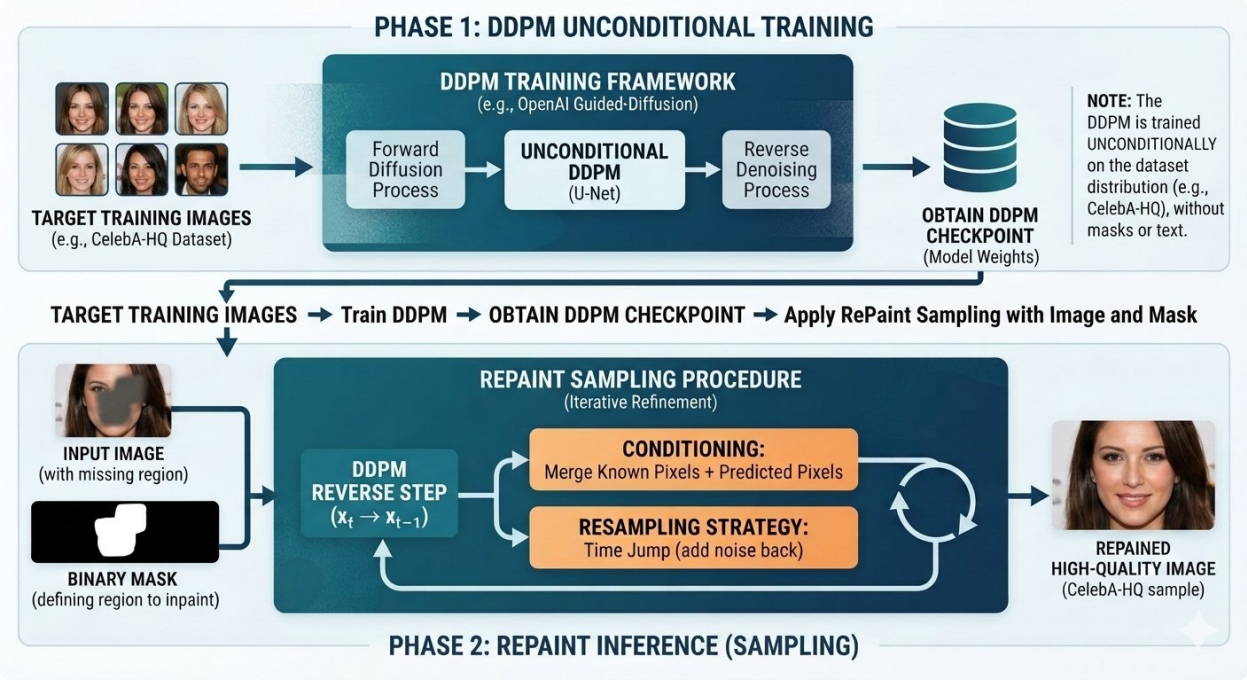
Limitations:

- Generation is mainly focused on RGB visual camouflage
- It does not directly handle 8-band multispectral image generation.
- Generated camouflage quality may depend on mask quality and object-background compatibility.
- Small objects may not be preserved clearly without crop-based processing.

- It may generate visually realistic images but does not guarantee that the object is spectrally camouflaged in non-RGB bands.
- For MCOD, extra adaptation is needed for multispectral consistency and small-object preservation.

Relevance to Present Work:
LAKE-RED is highly relevant because it is one of the most direct camouflage image generation baselines. It can be used first on MCOD official false-colour RGB images to generate camouflaged samples and test whether synthetic images can confuse or stress COD models. However, for the present MCOD project, LAKE-RED should be treated as an RGB/false-colour baseline. Since MCOD contains 8-band multispectral data and many small objects, further improvements such as crop-based generation, mask refinement, and spectral reconstruction or spectral matching are required before it can become a complete multispectral camouflage generation solution.

5. Lugmayr et al. (2022) — RePaint: Inpainting Using Denoising Diffusion Probabilistic Models





Review:

RePaint is a diffusion-based image inpainting framework rather than a camouflage image generation model. Its objective is to realistically reconstruct masked regions of an image while preserving the surrounding known content. Unlike conventional inpainting methods, RePaint does not train a dedicated mask-conditioned network. Instead, it performs a modified reverse diffusion process using a pretrained unconditional diffusion model, allowing it to generate visually consistent content for arbitrary mask shapes. The repeated resampling strategy helps improve the continuity between generated and known regions, producing seamless inpainting results.

Methods Used:

RePaint is built on a pretrained **Denoising Diffusion Probabilistic Model (DDPM)**. During inference, the known image region is repeatedly reintroduced into the reverse diffusion process while the unknown region is generated by the diffusion model. A **resampling schedule** alternates between forward and reverse diffusion steps, allowing the generated region and the surrounding image to better harmonize. Unlike text-guided diffusion models, RePaint relies only on the input image, binary mask, and pretrained diffusion prior. Importantly, RePaint **does not train its own diffusion model**; instead, it requires a separately trained unconditional DDPM (such as ImageNet, CelebA-HQ, or a custom-trained model) before the RePaint sampling algorithm can be applied.

Advantages:

- Does not require training a dedicated mask-conditioned inpainting network.
- Works with arbitrary and irregular mask shapes.
- Preserves the original image outside the masked region.
- Produces realistic image completion using diffusion priors.
- Publicly available implementation with pretrained checkpoints.
- Can be adapted to new datasets by replacing the pretrained diffusion model.

Limitations:

- Requires a pretrained unconditional diffusion model before RePaint can be used.
- For domain-specific datasets (such as LAKE-RED or camouflage datasets), a **new diffusion model must first be trained** on the target dataset.
- Does not use text prompts or semantic guidance for controllable generation.
- Generation is slower due to repeated resampling during inference.
- Original implementation mainly supports RGB images and is not directly applicable to 8-band multispectral data.
- Generated content may not preserve object semantics when the masked region is very large or contains complex structures.
- Performance may degrade for very small camouflage objects because no explicit object-aware mechanism is used.

Relevance to Present Work:

RePaint is relevant because it provides a strong **diffusion-based image inpainting baseline** for camouflage image generation. However, unlike CT-CIG or LAKE-RED, RePaint is **not an end-to-end trainable camouflage generation framework**. To use it on the LAKE-RED dataset, an unconditional diffusion model must first be trained on LAKE-RED images, after which RePaint can use the trained checkpoint for mask-guided image generation. For the MCOD dataset, additional modifications would be required to support 8-band multispectral images, small-object-aware generation, and spectral consistency. Therefore, RePaint serves as a useful baseline for evaluating general diffusion-based inpainting but requires significant adaptation before it can be effectively applied to camouflage image generation in multispectral settings.

Research Gap

- Existing methods are mostly RGB-focused and do not fully exploit 8-band multispectral information
- It is unclear whether improvements come from genuinely useful spectral information or only from changes in initialization, preprocessing, or training configuration.
- MCOD has many tiny targets, and both detection and generation models can miss, blur, or ignore small masked regions.
- A target may look camouflaged in false RGB but still remain detectable in hidden MSI bands.
- Most camouflage-generation and detection methods mainly focus on RGB images but reduce the performance when used with false rgb images.
- Detection models are rarely evaluated under generated camouflage/spoofing conditions
- The potential of synthetic camouflage generation for data augmentation and model improvement remains largely unexplored.
- Detection models may generate images which are unrealistic according to any real scenario.
- If they produce real images then they does not follow the ground truth's position size and other characteristics.
- Many generation methods limit for very large or very small objects

Work Completed

- Studied the MCOD dataset and verified image, mask and 8-band data structure.
- Manifested false-rgb-images form multispectral view
- Trained and evaluated SINet-V2, CamoFormer and ZoomNet on false-colour and 8-band inputs.
- Selected trained COD models for evaluating generated camouflage images.
- Reviewed recent camouflage image-generation methods.
- Selected CT-CIG as the first generation model.
- Literature review for the CT-CIG model is done.

- Prepared the CT-CIG notebook workflow and dataset structure.
- Started prompt generation and model training.
- Used Qwen-3.5-VL-3b model and changed the instruction prompts for rgb MCOD specific data later planning to do prompt adaptation for multispectral data
- Trained and tested the CT-CIG model on different settings 5-6 times to look for the result improvement
- Selected LAKE-RED as second generation model and literature review is also completed.
- Using data mentioned in paper, data consistency and sanity checks are done .
- Made pipeline reproducible for training and testing ,trial 1 epoch run in also done before running the final training to check if checkpoints are being saved and no NAN values appearing for loss during training.
- Trained the LAKE-RED style data for CT-CIG model with prompts and all as mentioned in original CT-CIG's paper
- Testing and evaluation is done for .

Results

Detection:

Final Comparison Table

Rank	Model	Input	MAE ↓	Dice ↑	mIoU ↑	Boundary-F ↑	S-measure ↑	W-Fmeasure ↑	Main Observation
1	**CamoFormer**	**MSI-8**	**0.00320**	**0.7124**	**0.5936**	**0.5594**	**0.8332**	**0.6905**	Best overall performance
2	SINet-V2	False-colour RGB	0.00425	0.6301	0.5096	0.4382	0.7893	0.5929	Strong RGB baseline
3	SINet-V2	MSI-8	0.00405	0.6257	0.5066	0.4418	0.7893	0.5881	Stable MSI-8 result
4	CamoFormer	False-colour RGB	0.00353	0.5935	0.4777	0.4460	0.7589	0.5929	Good precision, lower recall
5	ZoomNet	False-colour RGB	0.00448	0.4078	0.3197	0.3316	0.6773	0.4110	Moderate performance
6	ZoomNet	MSI-8	0.00450	0.3902	0.3111	0.3115	0.6742	0.3955	MSI-8 did not improve
7	CamoFormer	Scratch	0.00685	0.2110	0.1562	0.1146	0.5914	0.2141	Weak without pretraining

ZoomNet run	Input	S-measure ↑	wF-measure ↑	MAE ↓	adpEm ↑	meanEm ↑	maxEm ↑	adpFm ↑	meanFm ↑	maxFm ↑
ZoomNet_FC	False-RGB / false-colour 3ch	0.707	0.438	**0.003**	0.749	0.671	0.749	0.459	0.472	0.489
ZoomNet_MSI8 old init	8-band	0.674	0.396	0.00449	0.716	0.610	—	0.479	0.440	—
ZoomNet_MSI8 new init	8-band	**0.785**	**0.595**	**0.003**	**0.813**	**0.765**	**0.818**	**0.605**	**0.629**	**0.649**

Result for Zoomnet was improved for a better initialization.

Conclusion from results

The evaluated models were initialized using the official checkpoints released by the original authors and subsequently fine-tuned on the MCOD dataset. Therefore, the obtained results represent adapted and fine-tuned versions of the original architectures rather than direct reproductions of the benchmark baselines.

ZoomNet and CamoFormer benefit substantially from multispectral input, whereas SINet-V2 shows

almost no improvement, also results for Zoomnet is improved for a better initialization , suggesting that architectural design and band initialization strongly influences the

Ability to exploit spectral information.

Comparison from existing Baselines

Method	Input	E ↑	S ↑	F ↑	MAE ↓	Note
SINet	RGB	0.695	0.601	0.335	0.005	Existing MCOD baseline
SINet	MSI	0.758	0.616	0.369	0.006	Existing MCOD baseline
CODCEF	RGB	0.718	0.632	0.359	0.005	Existing MCOD baseline
CODCEF	MSI	0.763	0.677	0.444	0.004	Existing MCOD baseline
C2FNet-V2	RGB	0.885	0.743	0.553	0.009	Existing strong baseline
C2FNet-V2	MSI	0.913	0.810	0.654	0.008	Existing strong MSI baseline

PCNet RGB 0.397 0.788 0.149 0.005 Existing MCOD baseline

PCNet MSI 0.633 0.855 0.386 0.003 Existing MCOD baseline

SINet-V2 False RGB **0.845** **0.630** **0.555** **0.004** Ours, fine-tuned

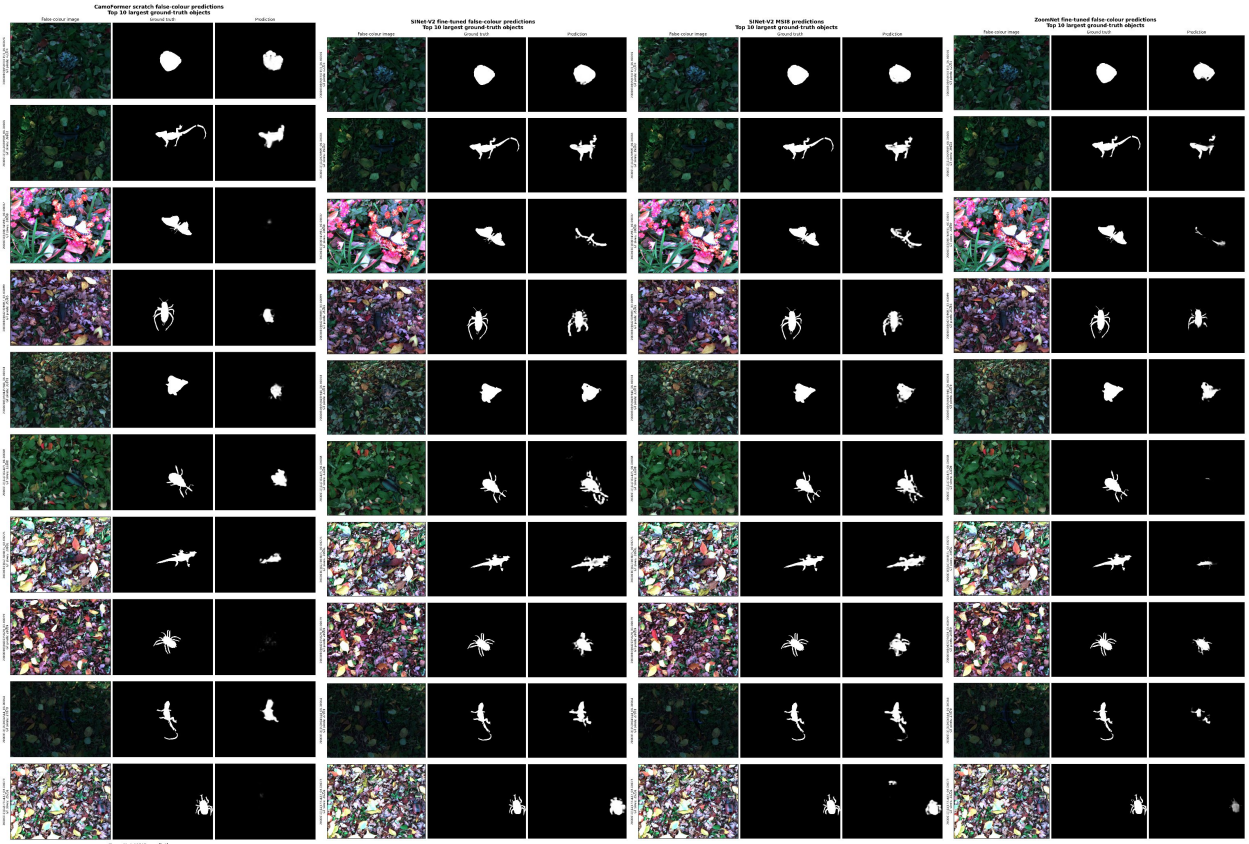
SINet-V2 MSI8 **0.848** **0.626** **0.552** **0.004** Ours, fine-tuned

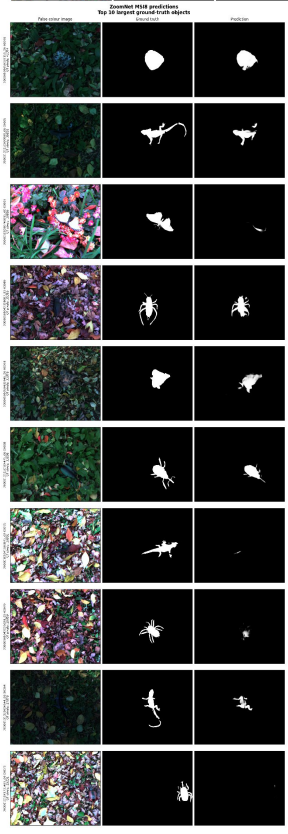
CamoFormer False RGB **0.903** **0.594** **0.642** **0.0035** Ours, fine-tuned

CamoFormer MSI8 **0.921** **0.712** **0.690** **0.0032** Ours, fine-tuned

ZoomNet False RGB **0.749** **0.707** **0.489** **0.003** Ours, fine-tuned

ZoomNet MSI8 **0.818** **0.785** **0.649** **0.003** Ours, fine-tuned


CamoFormer scratch false-colour predictions
Top 10 largest ground-truth objects
False-colour image
Ground truth
Prediction
SINet-V2 fine-tuned false-colour predictions
Top 10 largest ground-truth objects
SINet-V2 MSIR predictions
Top 10 largest ground-truth objects
Zoomtlot fine-tuned false-colour predictions
Top 10 largest ground-truth objects


Zoomtlot MSIR predictions
Top 10 largest ground-truth objects
False-colour image
Ground truth
Prediction

++

Generation:

COD generation plan:::

We use the LAKE-RED dataset protocol for model comparison. The training set contains 4,040 camouflaged images from COD10K and CAMO. For testing, we use three subsets: COD/CO, SOD/SO, and SEG/GO, each containing 6,473 image-mask pairs, resulting in 19,419 test inputs. Each model generates camouflaged images from the same test inputs. Following LAKE-RED, we compute FID, KID, LISPS, PSNR and l1 score between generated images and the real COD benchmark reference set COD10K_CAM. Results are reported separately for COD, SOD, SEG, and Overall. To ensure fairness, all models use the same input resolution, masks, random seeds, output count, and evaluation script.**Models selected and common dataset summary and Input requirements**

Model	Paper / year	Type	Input needed	Code availability	Use with LAKE-RED data	Recommendation
LAKE-RED	*LAKE-RED: Camouflaged Images Generation by Latent Background Knowledge Retrieval-Augmented Diffusion*, CVPR 2024	Camouflage-specific latent diffusion	Foreground image + mask	**Official code available**	Directly uses LAKE-RED dataset	**Must use as main baseline**
CT-CIG	*Text-guided Controllable Diffusion for Realistic Camouflage Images Generation*, AAAI 2026 / arXiv 2025	Text-guided camouflage diffusion	Image + mask + prompt	**Official GitHub available**	Can be adapted to LAKE-RED images/masks by generating prompts	**Strongest modern comparison**
LDM Inpainting	*High-Resolution Image Synthesis with Latent Diffusion Models*, CVPR 2022	Generic latent diffusion inpainting	Image + mask	**Official code available**	Directly compatible with LAKE-RED-style image/mask input	**Good generic diffusion baseline**
RePaint / RePaint-L	*RePaint: Inpainting using Denoising Diffusion Probabilistic Models*, CVPR	Diffusion inpainting	Image + mask	**Official code available**	Compatible after formatting masks/images	**Good inpainting baseline**

Split	Source	Count	Use
Training	COD10K + CAMO	**4,040**	Train/fine-tune models
COD / CO test	CAMO + COD10K + NC4K	**6,473**	Camouflaged-object test input
SOD / SO test	Salient-object datasets such as DUTS / DUT-OMRON	**6,473**	Salient-object test input
SEG / GO test	General segmentation datasets such as COCO2017	**6,473**	General-object test input
Overall test	COD + SOD + SEG	**19,419**	Full evaluation
Reference real set	`COD10K_CAM` in repo	**5,066** in your local repo	Real distribution for FID/KID

LAKE-RED	`image + mask`
CT-CIG	`image + mask + generated prompt`
LDM Inpainting	`image + mask`
RePaint-L	`image + mask`

3 models are implemented till now

1 CT-CIG model (batch size 4)

Prompts are trained on a Qwen2.5-VL 3b parameter model(prompts may not be richer but training time is reduced)

Sample prompt:`"detail_prompt": "A small, irregularly shaped rock lies on a patch of dry grass and scattered fallen leaves. The rock has a greenish-brown color and a rough texture. It blends seamlessly with the surrounding environment, which features muted earth tones and a mix of textures. The lighting is natural, casting subtle shadows that enhance the rock's natural appearance.", "prompt": "A camouflaged rock blends into a dry grassy area with scattered leaves, showcasing natural textures and muted colors."`

Tried using prompt generation with Qwen2.5-VL 7b parameter model(richer prompt,better understanding of surrounding and object but taking so much time for prompt training.)

Sample prompt:`"detail_prompt": "The image shows a close-up view of a ground covered with dry grass and scattered dried leaves in various shades`

of brown and yellow. The grass appears somewhat sparse, with some areas showing more exposed soil. In the center of the frame, there is an object with a distinctive, irregular shape and a patterned surface resembling a mosaic or a piece of broken pottery, primarily in muted tones of gray, brown, and white. The object stands out slightly against the background due to its contrast in color and texture, yet it blends well enough with the surrounding dry leaves and grass to suggest some level of camouflage.", "prompt": "*Camouflaged object* A mosaic-patterned rock with muted tones in a grassy area with scattered dry leaves."

20 epoch setup: sampled at 3000th step /5132 steps

The model is able to learn the background pattern, natural camoflage and mask position target but cannot define the object shape and type. This is with the setup mentioned in paper

GROUND TRUTH GENERATED SAMPLES:1,2,3,4



10 epoch setup: sampled at 750/2570 steps with different setting of hyper parameters

The model is able to target the object and the background and also learns the camouflage but cannot target the mask position.



20 epoch setup: sampled at



1500/5132 steps with better prompt settings and higher control-net scale

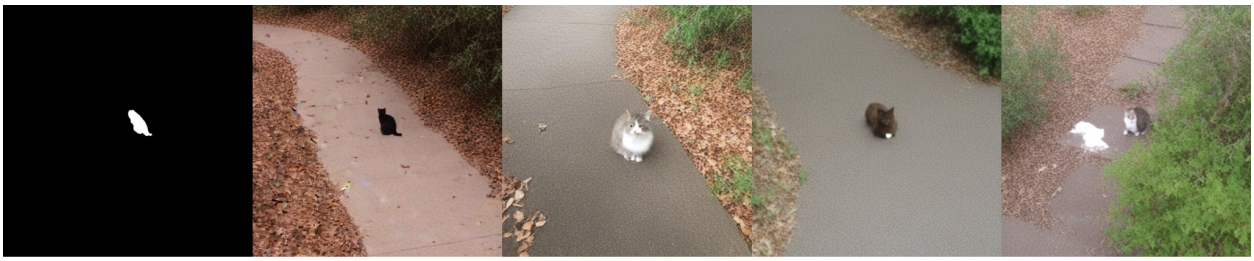
A more realistic image,model is able to learn the mask's position, less camouflage adaptation ,generating white blob like things.

20 epoch setup: sampled at 1500/5132 steps with better prompt settings higher learning rate and higher inference rate. Better camouflage ,but still not good enough to be used for testing.

.



Sample training:20 epoch setup



more realistic photoes but not camouflage

Problems faced when working with CT-CIG

- MCOD images have strong blue/cyan/green false-colour tones. But CT-CIG and SDXL are naturally biased toward real RGB photography.
- Prompts are making the trainig and generation too biased for object rather than following the mask.
- The model learns the background very well, but the target object signal is weak as many MCOD objects are smaller. After resizing to 512 x512 , the object may occupy a small number of pixels.
- Tried using 5-6 different setups but still not able to generate perfect results.
- Now planning to do different experimental setup for this model
- 7b parameter qwen prompt generation model is taking too much time for training and generating the prompts.

Results after using Qwen 7b_model_prompt for training

20 epoch setup: sampled at 5000th/5120 steps with lower FIRM loss rate

Improvement:

- Enhanced texture generation

- Better mask control
- No over fitting at the steps after 2500
- Learning camouflage



Metrics after testing:FID and KID

```
Frechet Inception Distance: 161.8639
Kernel Inception Distance: 0.07059401 ± 0.01011972
frechet_inception_distance: 161.8639
kernel_inception_distance_mean: 0.07059401
kernel_inception_distance_std: 0.01011972
```

Note:

1. FID measures the distance between the distribution of generated images and images of a particular set .

Formula

$$FID = \|\mu_r - \mu_g\|^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right)$$

where:

- μ_r = mean of real image features
- μ_g = mean of generated image features
- Σ_r = covariance of real features
- Σ_g = covariance of generated features

2. KID also compares the feature distributions of real and generated images, but uses **kernel methods** instead of fitting Gaussian distributions.

Formula

KID is based on the **Maximum Mean Discrepancy (MMD)** with a polynomial kernel.

Conceptually,

$$\mathrm{KID} = \mathrm{MMD}^2(\mathcal{X}, \mathcal{Y})$$

where

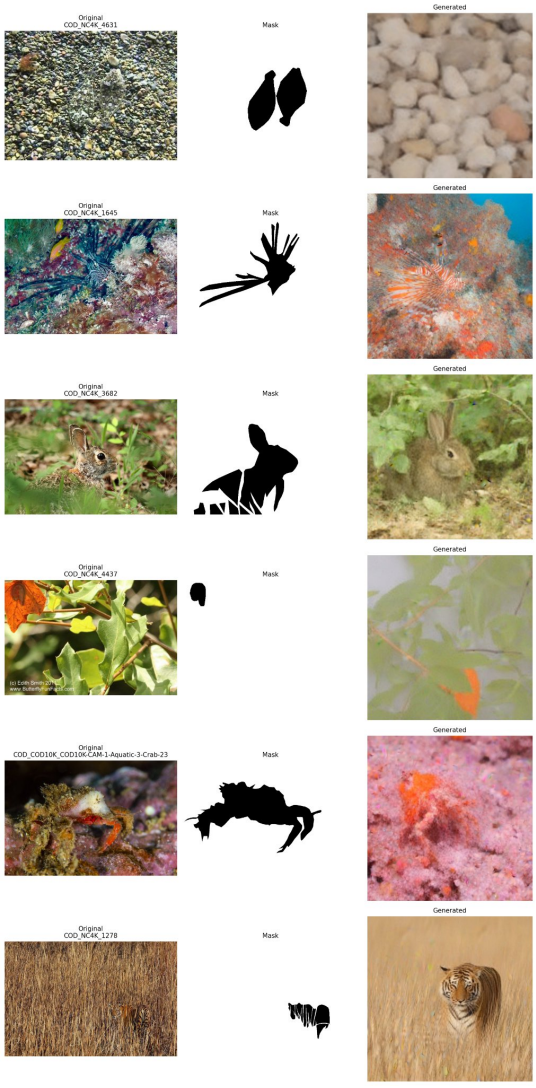
- $\mathcal{X}$ = real image features
- $\mathcal{Y}$ = generated image features

Final Results on some test images:


Original
19700101000130315_00-00000
Mask
Generated
Original
19700101000107639_00-00017
Mask
Generated
Original
19700101000131863_00-00018
Mask
Generated
Original
19700101000156031_00-00019
Mask
Generated
Original
19700101000139629_00-00021
Mask
Generated
Original
19700101000155666_00-00022
Mask
Generated
Original
19700101000124693_00-00024
Mask
Generated
Original
19700101000737518_00-00037
Mask
Generated
Original
19700101000752489_00-00038
Mask
Generated
Original
19700101000808362_00-00039
Mask
Generated

Training and testing on LAKE-RED style dataset(as mentioned on original paper)

COD



SOD


Original
SEG_COCO2017_000000460503
Mask
Generated
Original
SEG_COCO2017_000000245295
Mask
Generated
Original
SEG_COCO2017_000000388564
Mask
Generated
Original
SEG_COCO2017_000000450217
Mask
Generated
Original
SEG_COCO2017_000000051335
Mask
Generated
Original
SEG_COCO2017_000000222054
Mask
Generated
vadakkus

SEG(general objects)



METRICS

Full-image metrics

Category	PSNR ↑	LPIPS ↓	L1 error ↓
COD	10.1567	0.8741	0.270795
SOD	**10.2048**	**0.8216**	**0.263516**
SEG	9.7814	0.9126	0.276211

Background metrics

Category	PSNR ↑	LPIPS ↓	L1 error ↓
COD	10.0996	0.7373	0.275131
SOD	**10.3161**	**0.6398**	**0.264042**
SEG	9.8513	0.7334	0.277187

LAKE-RED FID/KID protocol

Category	FID ↓	KID ↓
COD	**56.5498**	**0.020100 ± 0.000746**
SOD	88.4814	0.025309 ± 0.000820
SEG	111.4552	0.031844 ± 0.001091

2 LAKE-RED model

Methods		Input	Camouflaged Objects		Salient Objects		General Objects		Overall	
			FID↓	KID↓	FID↓	KID↓	FID↓	KID↓	FID↓	KID↓
Image Blending	AB [10]$_{03}$	$\mathcal{F}+\mathcal{B}$	117.11	0.0645	126.78	0.0614	133.89	0.0645	120.21	0.0623
	CI [7]$_{10}$	$\mathcal{F}+\mathcal{B}$	124.49	0.0662	136.30	0.7380	137.19	0.0713	128.51	0.0693
	AdaIN [21]$_{17}$	$\mathcal{F}+\mathcal{B}$	125.16	0.0721	133.20	0.0702	136.93	0.0714	126.94	0.0703
	DCI [53]$_{20}$	$\mathcal{F}+\mathcal{B}$	130.21	0.0689	134.92	0.0665	137.99	0.0690	130.52	0.0673
	LCGNet [29]$_{22}$	$\mathcal{F}+\mathcal{B}$	129.80	0.0504	136.24	0.0597	132.64	**0.0548**	129.88	0.0550
Image Inpainting	TFill [64]$_{22}$	$\mathcal{F}$	63.74	0.0336	96.91	0.0453	122.44	0.0747	80.39	0.0438
	LDM [41]$_{22}$	$\mathcal{F}$	58.65	0.0380	107.38	0.0524	129.04	0.0748	84.48	0.0488
	RePaint-L [36]$_{22}$	$\mathcal{F}$	76.80	0.0459	114.96	0.0497	136.18	0.0686	96.14	0.0498
	Ours$_{23}$	$\mathcal{F}$	**39.55**	**0.0212**	**88.70**	**0.0428**	**102.67**	0.0625	**64.27**	**0.0355**

First method we are doing is to reproduce papers results by training from scratch on the datasets available as mentioned in the paper. And later adapting on our MCOD dataset.

Dataset used:

training 4,040 images (3,040 from COD10K, 1,000 from CAMO)

testing 6,473 pairs of Camouflaged Objects (CO), Salient Objects (SO), and General Objects (GO) each

Total::3 X 6473 =19419

```
Train Images: 4040
Train Masks : 4040
Val Images  : 19419
Val Masks   : 19419
```

```
Images: 4040
Masks : 4040
Missing Masks : 0
Missing Images: 0
```

```
Images: 19419
Masks : 19419
Missing Masks : 0
Missing Images: 0
```

Below are the major modules in the network and the number of trainable parameters in each.

```
  | Name              | Type             | Params
-------------------------------------------------------
0 | model             | DiffusionWrapper | 387 M
1 | model_ema         | LitEma           | 0
2 | first_stage_model | VQModelInterface | 53.2 M
-------------------------------------------------------
```

Experimental setup 1 (best 20 epoch)

4040 training images,200 images from validation for validation loss in every 5 epochs.

Optimization: Time for training a single epoch is reduced from 60 mins to 39 mins.

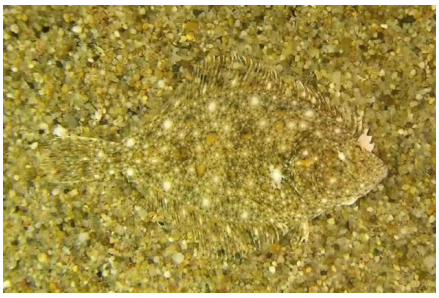
Got a training error at 26th epoch 1/jul.

Checked some samples at the best 20th epoch.

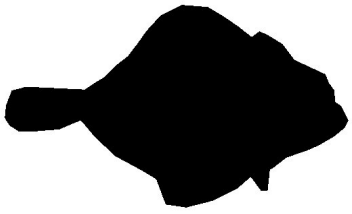
Tried retraining but no improvement in final loss.(best epoch 20th)

Changed to batch size 1 from 2 as error was popping ”out of memory”.

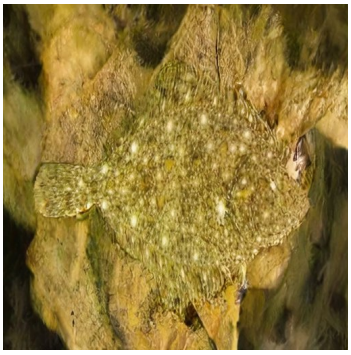
Did testing at 20th epoch’s checkpoint



Original



GT



generated

Checkpoint	Images	FID ↓	KID ↓
Epoch 19	200	150.8278	0.03118 ± 0.00214

Result for 200 COD generated images for this experiment

Note:actual value may differ if evaluated on full validation dataset

Why 200 only:testing/generation is taking so much time so partially evaluated for the first experiment

Reason for bad training: some optimization steps may affect the model learning and image background frequency generation, fixed in experiment 2 .Later found out initialization for the masks must be changed ,fixed in experiment 2,unrealistic background generation,fixed in experiment 2.

Experimental setup 2 :30 epoch setup(took approx 23 hrs**)**

Reason for 30 epoch:

- Validation stopped improving meaningfully after epoch 11.
- The best EMA result appeared around epoch 23.
- The learning rate reached its minimum around epoch 26.
- Training and reconstruction losses have plateaued.

metrics

Higher PSNR = generated region is closer to original

Lower LPIPS = perceptually closer to original

Lower masked L1 error = lower pixel difference

KID

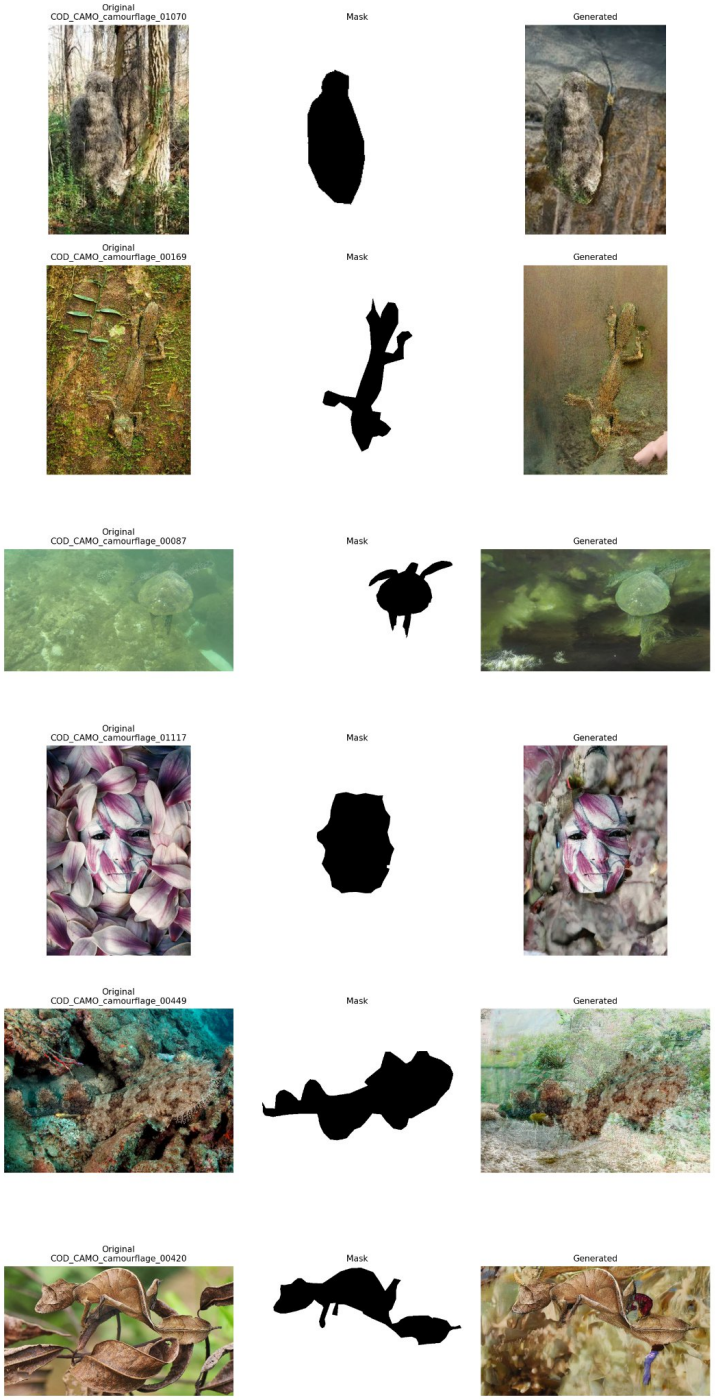
FID

Evaluation protocol

- Took 5,066 data/COD10K_CAM reference images for FID and KID calculation
- 299×299 resizing for FID/KID evaluation
- torch-fidelity from the Conda ldm environment
- The all-image evaluation computes PSNR, LPIPS, and L1 error across the complete generated image, capturing both intended object-region modifications and unintended background changes.
- The background-only evaluation restricts these metrics to pixels outside the target object mask, measuring how effectively the generation process preserves the surrounding scene.

- all-image scores reflect overall reconstruction similarity, whereas background-only scores specifically quantify non-target-region preservation.

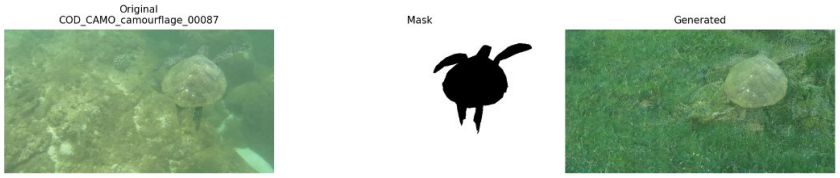
Results for best checkpoint(11th epoch)-for 205 test samples



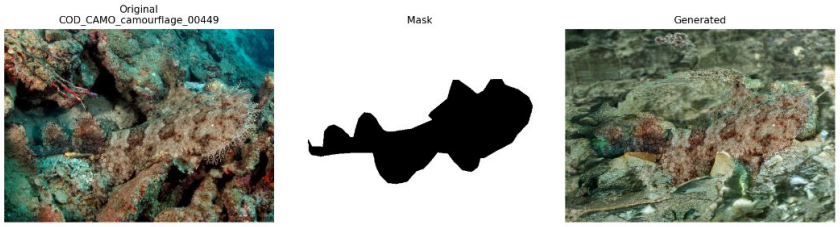
Results for last epoch(30th epoch)-for 200 samples

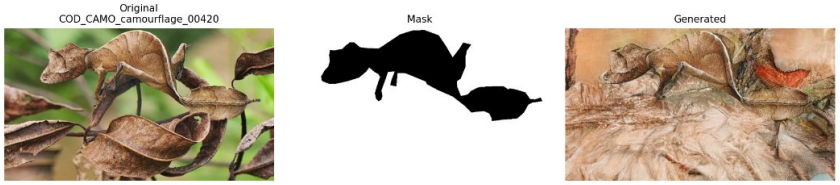












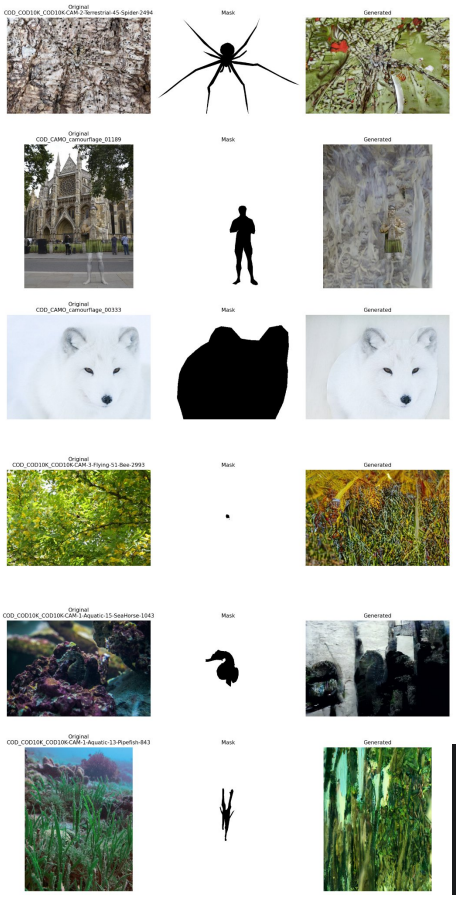
For the experimental tests ,epoch 30 gives the best results,therefore doing final testing and evaluation on the 30th epoch checkpoint for approx 200 generated images

Checkpoint	Images	FID ↓	KID ↓
Epoch 11	205	153.7184	0.03116 ± 0.00227
Epoch 30	200	**148.0902**	**0.02601 ± 0.00196**

FINAL evaluation RESULT:-

Testing is done on 2000 images out of 6473 evaluation images(approx 30% of the evaluation dataset)for every subset(COD,SOD,SEG)as running time was so much for the image generation task hence reduced the evaluation for faster results**.**

For Camouflage object subset(4hrs)-2000images COD



FID ↓	KID ↓
52.2279	0.02621 ± 0.00123

original,mask,generated

For salient object subset(4hrs)-2000images SOD

FID ↓	KID ↓
114.1423	0.05368377 ± 0.001814708





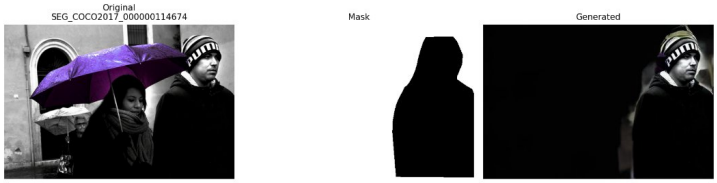




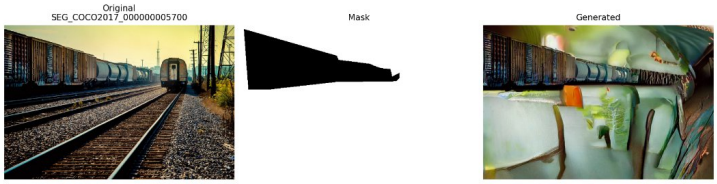




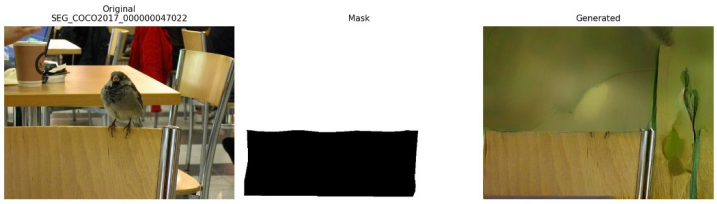
For general object dataset(4hrs)-2000images SEG

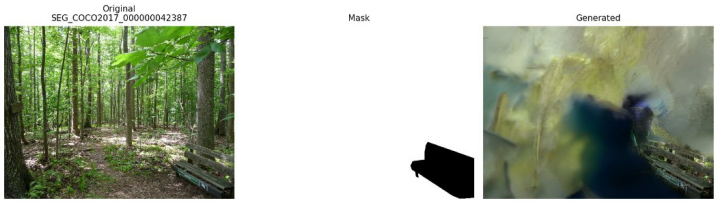












FID ↓	KID ↓
114.4261	0.06135229 ± 0.002122133

Conclusion

Dataset	Region	PSNR ↑	LPIPS ↓	L1 error ↓
COD	All image	11.16	0.7206	0.2251
COD	Background	19.71	0.0365	0.0825
SOD	All image	10.74	0.6435	0.2347
SOD	Background	**26.02**	**0.0128**	**0.0365**
SEG	All image	10.36	0.6955	0.2414
SEG	Background	23.41	0.0222	0.0503

Background-only evaluation produced substantially higher PSNR and lower LPIPS and L1 error than whole-image evaluation across all datasets. This indicates that most image changes are concentrated within the intended object region, while the surrounding background is comparatively well preserved. SOD achieved the strongest background-preservation performance, followed by SEG and COD. Whole-image similarity was lower because these scores include the intentionally modified object region; therefore, these results should not be interpreted solely as reconstruction failure. Overall, the model demonstrates effective localization of generation, although background preservation is comparatively weaker for COD samples.

3:RePaint

For training: Official RePaint itself does not train the model. For training, we trained an unconditional diffusion model on our LAKE-RED images, then used RePaint sampling during inference**.**

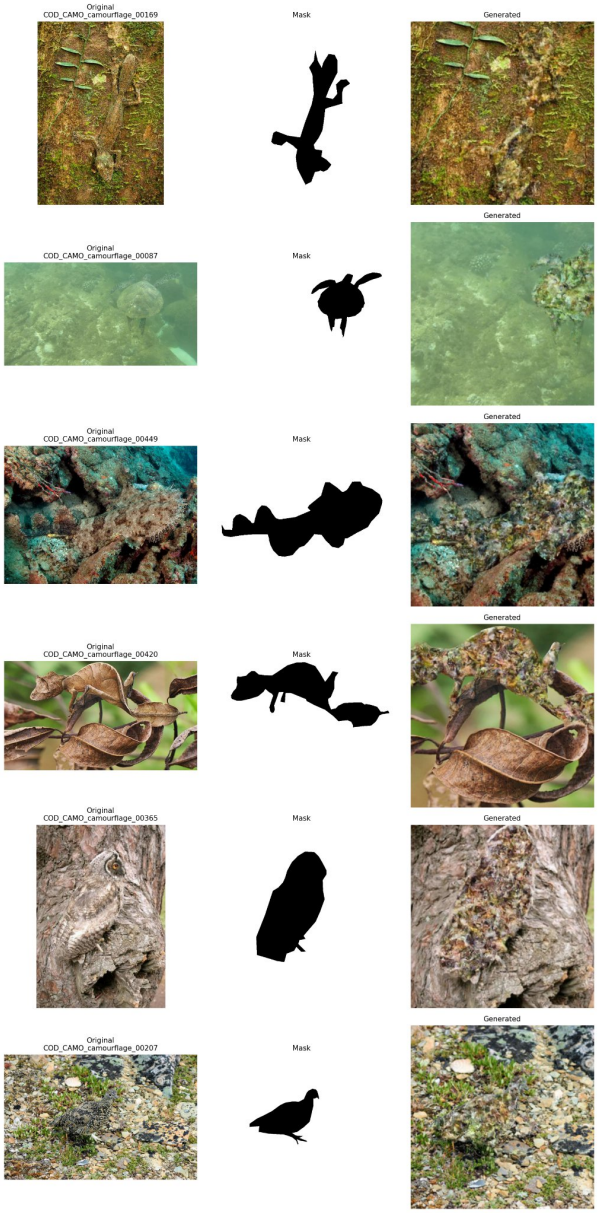
Each checkpoint in saved in every 5000th step.

No Training for a fixed epoch for unconditional diffusion model,so have to monitor the training and stopped at 20,000 th step (10 epoch) and this is our first experimental setup

Note: (background-white,object-black)while training masks should be like this

1st setup 20ksteps(approx 10 epochs)

Objects are understanding the background color and pattern but not consistent and realistic , maybe because of undertraining.Generated 140 images for initial testing stage. 5420 denoising/resampling



iterations.

.

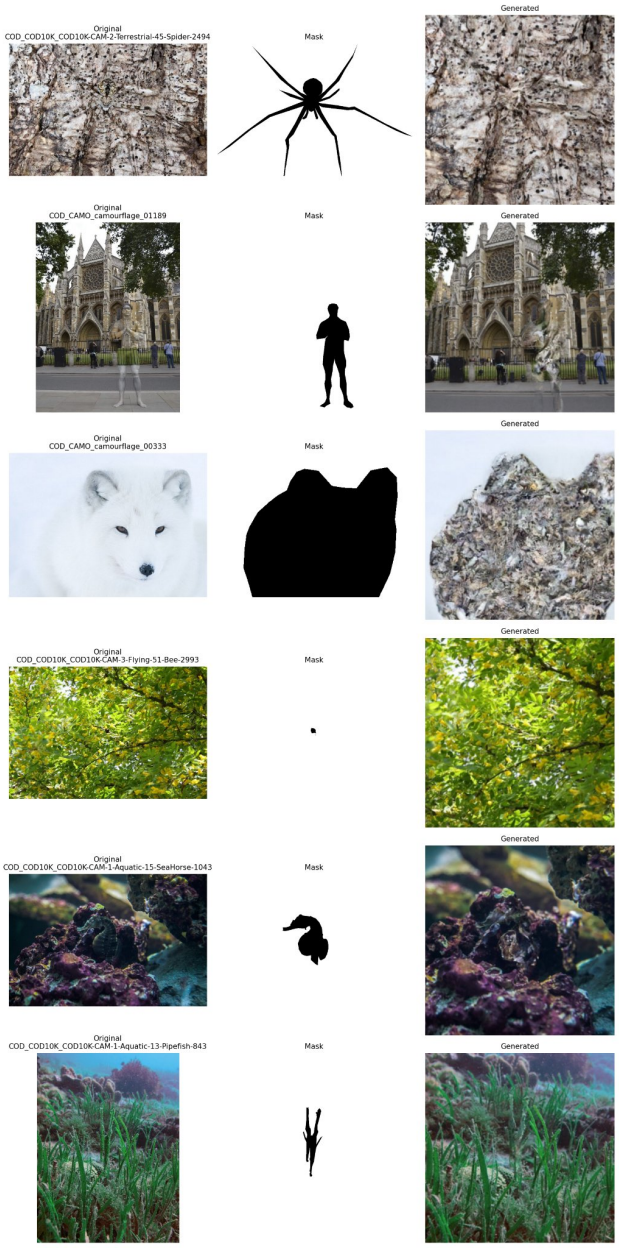
“Hole” means the inpainted/object-removed region

2nd setup:40k step training

N	PSNR Full ↑	L1 Full ↓	PSNR Hole ↑	L1 Hole ↓
140	18.6060	0.045500	12.4408	0.183560

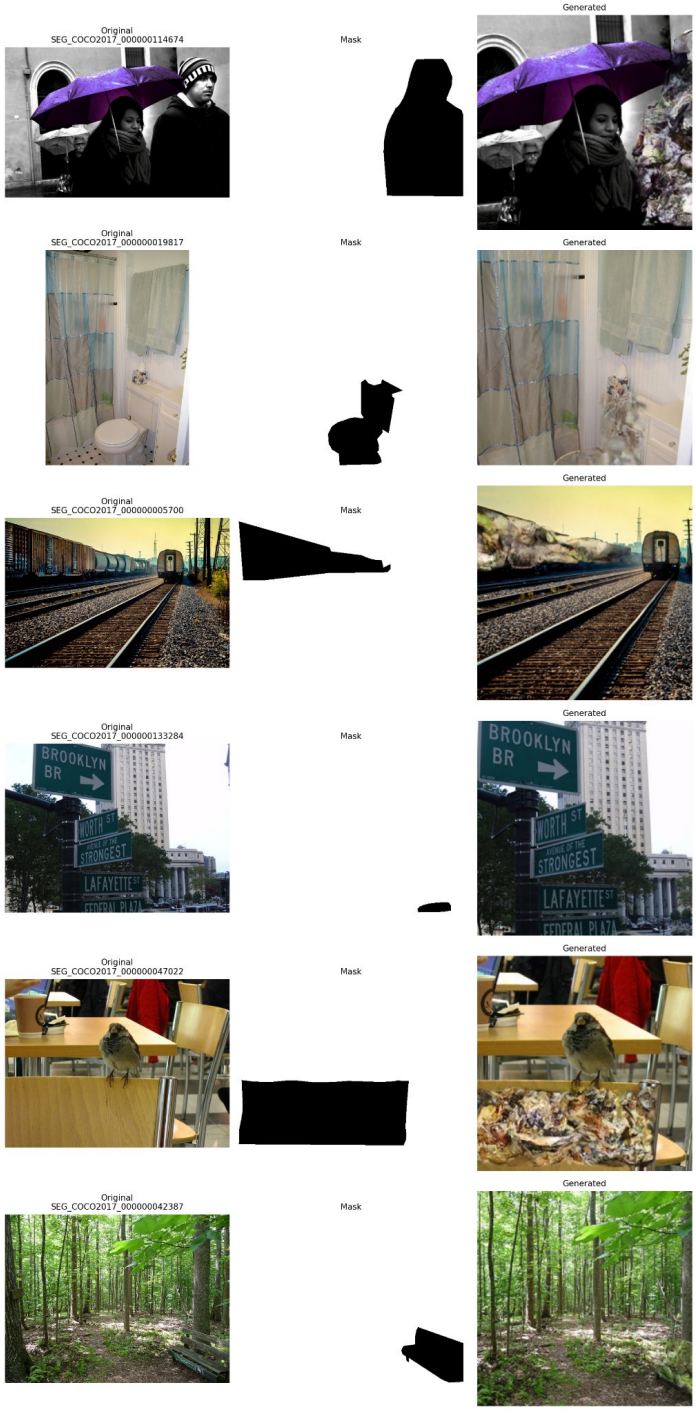
Distortion in the image is because 1:denoising steps are changed from 5420 to 990 to get results quickly .2:training was done for less number of steps

Here denoising/resampling iterations is reduced from 5420 to 990 and batchsize 8 to make training faster



COD

.


Original
SEG_COCO2017_000000114674
Mask
Generated
Original
SEG_COCO2017_000000019817
Mask
Generated
Original
SEG_COCO2017_000000005700
Mask
Generated
Original
SEG_COCO2017_000000133284
Mask
Generated
BROOKLYN BR
NORTH ST
AVENUE OF THE STRONGEST
LAFAYETTE ST
FEDERAL PLAZA
Original
SEG_COCO2017_000000047022
Mask
Generated
Original
SEG_COCO2017_000000042387
Mask
Generated

SOD

Original
SOD_DUTO_sun_bocaxozrgmudnwlf



Mask



Generated



Original
SOD_DUTO_sun_afnmfbppdjaptjiv



Mask



Generated



Original
SOD_DUTO_sun_abijsfnjgedupbvu



Mask



Generated



Original
SOD_DUTO_sun_bzdngjbkhqqvthyr



Mask



Generated



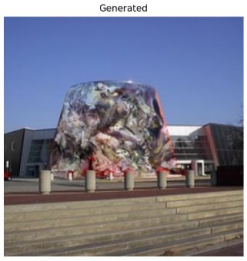
Original
SOD_DUTO_sun_anggafmyxweespqt



Mask



Generated



Original
SOD_DUTO_sun_almrdmqphdlabxsv



Mask



Generated



SEG

METRICS

Split	N	FID ↓	KID Mean ↓	KID Std
COD	2000	51.2874	0.014624	0.000784
SEG	2000	59.7114	0.023493	0.001106
SOD	2000	62.7121	0.021052	0.001139

Split	N	PSNR Full ↑	L1 Full ↓	PSNR Background ↑	L1 Background ↓	PSNR Hole ↓/↑	L1 Hole ↓	LPIPS ↓
ALL	6000	13.2672	0.140541	14.1258	0.122159	10.2640	0.241837	0.363078
COD	2000	13.3622	0.149167	13.6273	0.142636	11.9167	0.195126	0.353418
SEG	2000	12.6300	0.148872	13.6076	0.125290	9.5541	0.266174	0.393501
SOD	2000	13.9041	0.123583	15.3656	0.098551	9.6990	0.264211	0.342316

On full dataset

Metric	Value
FID ↓	35.1596
KID mean ↓	0.011982
KID std	0.000845

decoy-diff(proposed)

Note: we took pretrained RGB checkpoints for the detection models

Experimental setup 1

Trained the model on 20,000 steps(approx 20 epochs)

if original images have different resolutions, the model receives:

image: [B, 3, 512, 512]

mask: [B, 1, 512, 512]

problem:. The repeated "green/black striped" or "scale-like" texture appearing on completely different objects is a strong clue that the model has converged to an adversarial shortcut rather than learning camouflage**.**



Evidence from images

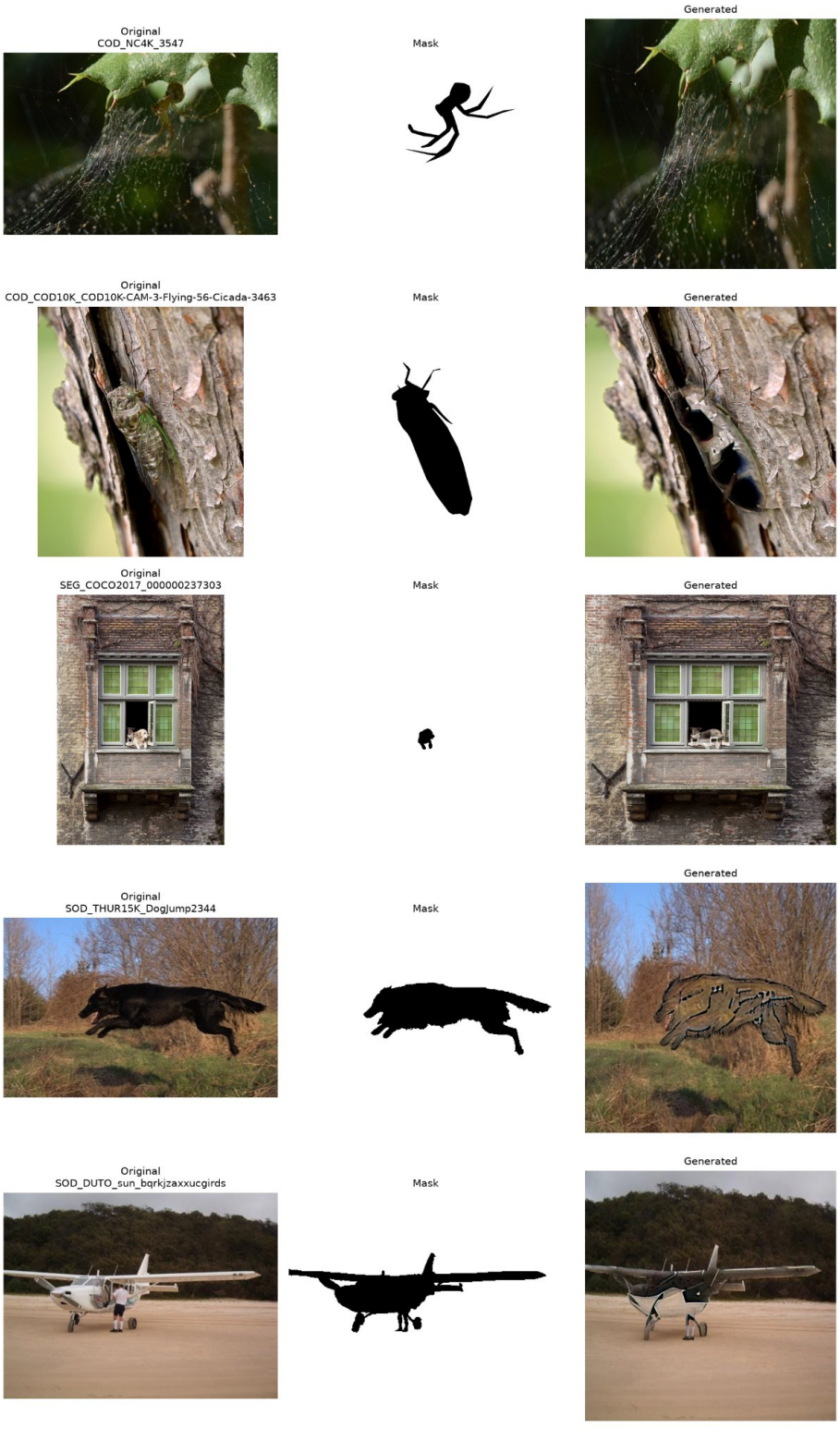
- Mask localization is correct .
- Background preservation is good.
- Every object gets the SAME texture .
- Model is learning general pattern, but not learning to learn local texture

Experimental setup 2

Changes done

- Add a local camouflage matching loss.
- Warm-start detector loss instead of using it from step 1.
- Make frequency loss local rather than global.
- Strengthen perceptual realism inside the object, not edge loss..
- Eventually replace
- CLIP style retrieval with retrieval codebook like LAKE-RED.

Result after changes



Experimental Setup 3

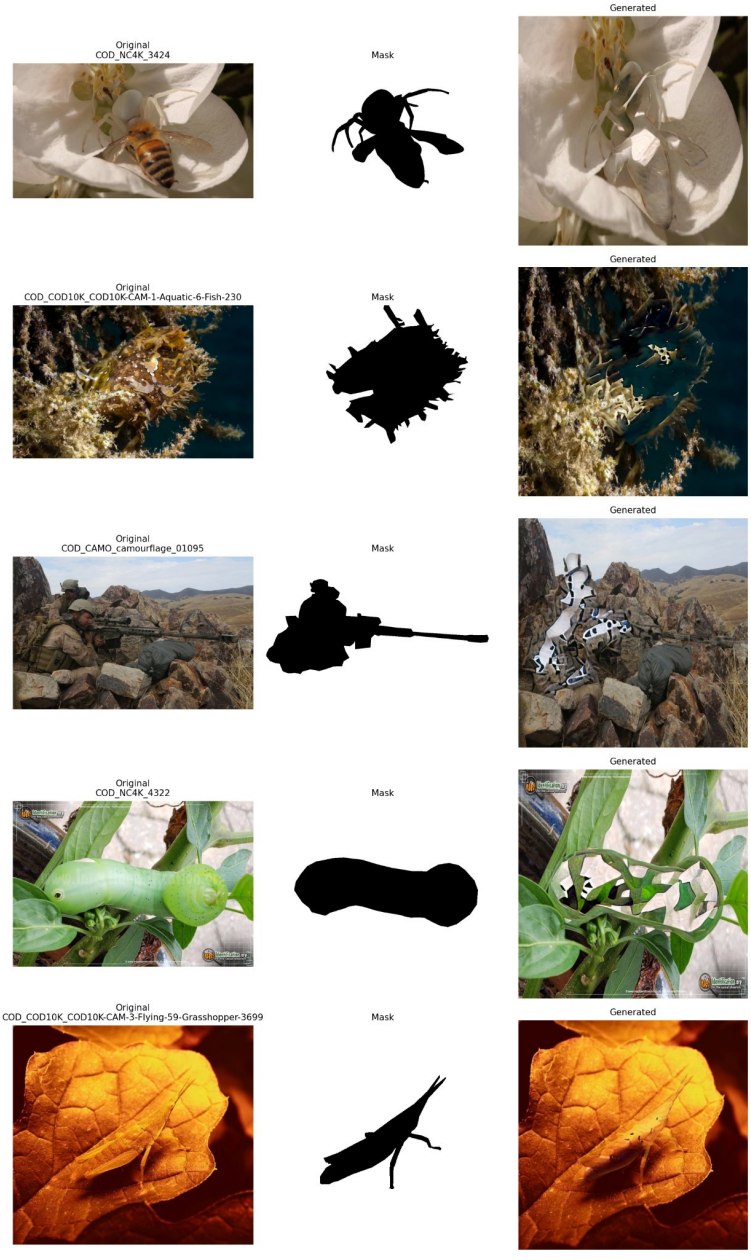
- The previous model preserved masks/background but produced broken patch artifacts inside large objects.
- This indicated over-strong local texture matching, not a denoising-step issue.
- lambda_camo was reduced from 1.0 to 0.5 to avoid forcing the whole object to copy nearby background texture.
- camo_texture_weight was reduced from 0.5 to 0.25 to reduce mosaic/pasted texture artifacts.
- lambda_freq was reduced from 0.1 to 0.05 because object-to-ring frequency matching was too strong for large object masks.
- lambda_obj was increased from 0.5 to 0.75 to preserve object structure and identity better.
- camo_tv_weight was increased from 0.1 to 0.2 to suppress high-frequency broken black/white patches.
- Training is resumed from the latest checkpoint for 5000 more steps instead of training from scratch.

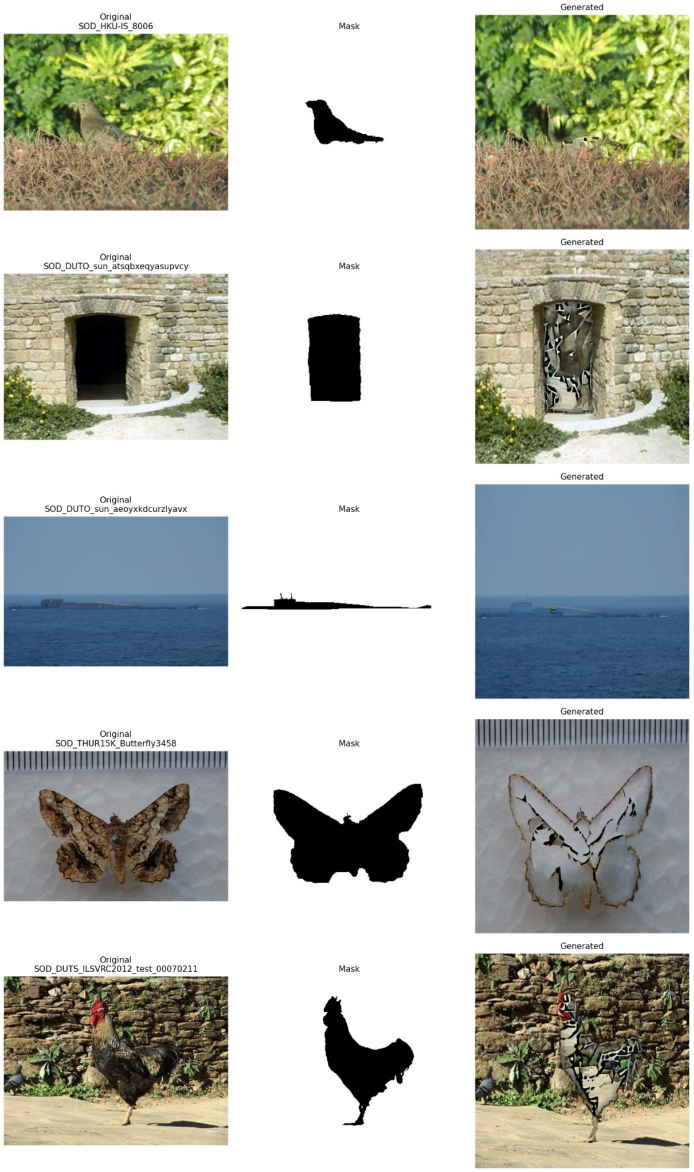
We got the best result (visually at least) for the experiment 2 for 20000th step run among 5 sample runs

Final result (on settings of experiment 2)

COD

```
========== RESULTS ==========
num_pairs    : 2000
PSNR         : 21.750573511600493
L1           : 0.04233787014905829
LPIPS        : 0.212530840144027
FID          : 36.76683807373047
KID_mean     : 0.009360904805362225
KID_std      : 0.002834958257153630
==============================
```

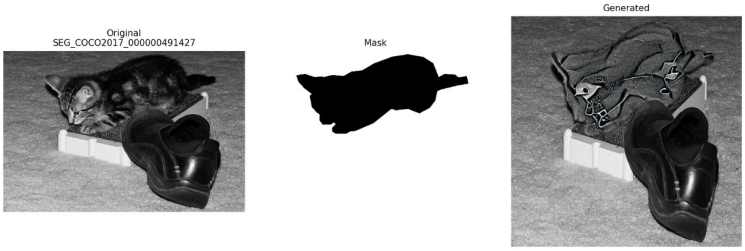




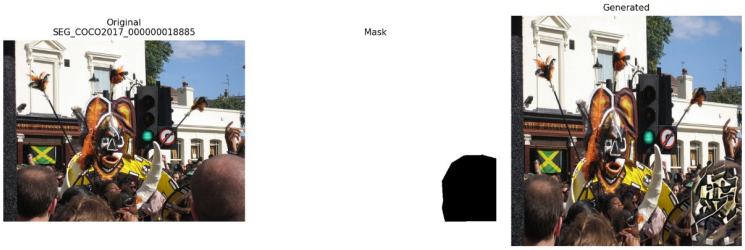
SOD

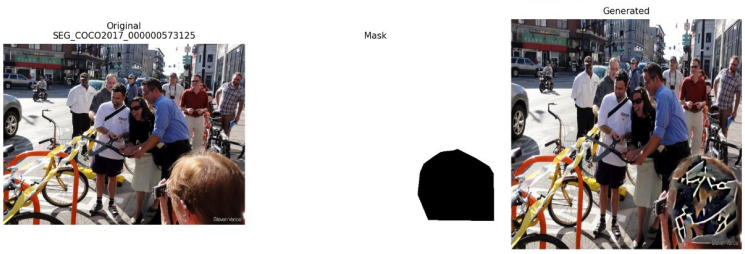
```
========== RESULTS ==========
num_pairs   : 2000
PSNR        : 18.898571059942245
L1          : 0.0552678855664562
LPIPS       : 0.1706137801809236
FID         : 109.83326721191406
KID_mean    : 0.0445504672825336
KID_std     : 0.0058673662133514
=============================
```

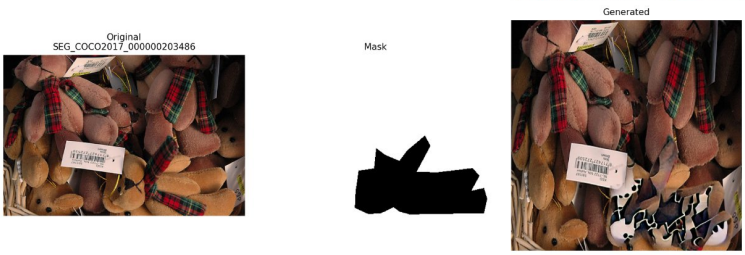
SEG












```
========== RESULTS ==========
num_pairs    : 2000
PSNR         : 18.913636793613435
L1           : 0.063834381158580008
LPIPS        : 0.2153245526445098
FID          : 144.52005004882812
KID_mean     : 0.058608561754226685
KID_std      : 0.006256142631173134
==============================
```

Scope of improvement

- Results may improve if used with more camouflage specific prompt
- Repetitive pattern in generated images may be fixed by tuning the hyperparameters
- For detector loss some other better detector for targeted images may be used to compare performance of the model .
- In some cases the model is not able to blend the boundary with the background,and could be fixed by some boundary aware loss.
- For very large object or complex background repetitive pattern occurs while generation
- May give better results on military dataset as visual notebooks in the architecture stores region specific patterns which is good for region specific dataset.

Model comparison

Training images : 4040

testing images : 6000 (2000 for each` set)

Models Compared

1. **Decoy-Diff (Proposed)**
2. **RePaint**
3. **LakeRed**
4. **CTCIG**

Comparing model category wise

1. COD Dataset Comparison

Model	PSNR ↑	L1 ↓	LPIPS ↓	FID ↓	KID ↓
Decoy-Diff	**21.75**	**0.0423**	**0.2125**	**36.77**	**0.0094**
RePaint	13.36	0.1492	0.3534	**51.29**	**0.0146**
LakeRed	11.16	0.2251	0.7206	52.23	0.0262
CTCIG	10.16	0.2708	0.8741	56.55	0.0201

Analysis

:The proposed Decoy-Diff model clearly dominates the COD benchmark. Compared to the closest competing model (RePaint), it achieves an 8.4 dB improvement in PSNR, a 72% reduction in L1 error, a

substantially lower LPIPS score, and the best FID and KID values. These results indicate superior reconstruction fidelity, perceptual similarity, and distributional alignment with real camouflaged images.

2. SOD Dataset Comparison

Model	PSNR ↑	L1 ↓	LPIPS ↓	FID ↓	KID ↓
Decoy-Diff	**18.90**	**0.0553**	**0.1706**	109.83	0.0446
RePaint	13.90	0.1236	0.3423	**62.71**	**0.0211**
LakeRed	10.74	0.2347	0.6435	114.14	0.0537
CTCIG	10.20	0.2635	0.8216	88.48	0.0253

Analysis:

On the SOD dataset, Decoy-Diff again provides the strongest reconstruction quality, with the highest PSNR and the lowest L1 and LPIPS values. However, RePaint achieves the best FID and KID, indicating that its generated images better match the statistical distribution of real SOD images. CTCIG ranks second in terms of distributional metrics but performs poorly in reconstruction quality, suggesting that it generates visually plausible images without accurately preserving scene content.

3. SEG Dataset Comparison

Model	PSNR ↑	L1 ↓	LPIPS ↓	FID ↓	KID ↓
Decoy-Diff	**18.91**	**0.0638**	**0.2153**	144.52	0.0586
RePaint	12.63	0.1489	0.3935	**59.71**	**0.0235**
LakeRed	10.36	0.2414	0.6955	114.43	0.0614
CTCIG	9.78	0.2762	0.9126	111.46	0.0318

Analysis:

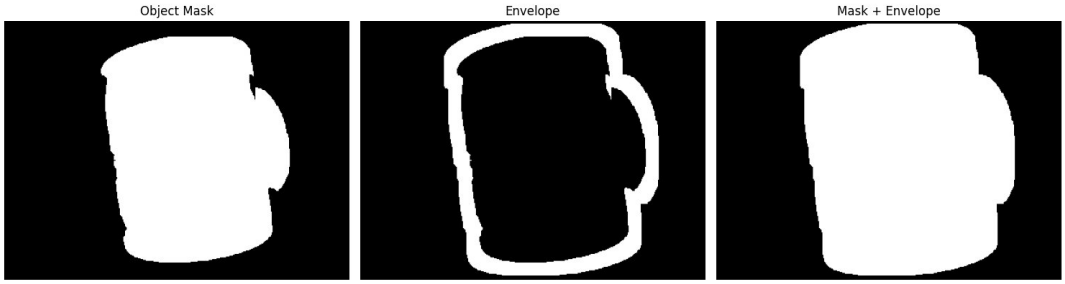
The SEG dataset remains the most challenging. Decoy-Diff maintains a clear lead in reconstruction fidelity, whereas RePaint produces outputs with the best FID and KID. CTCIG surpasses LakeRed in distributional metrics but still exhibits poor reconstruction quality, indicating limited preservation of segmentation structures.

METRIC (exploration)

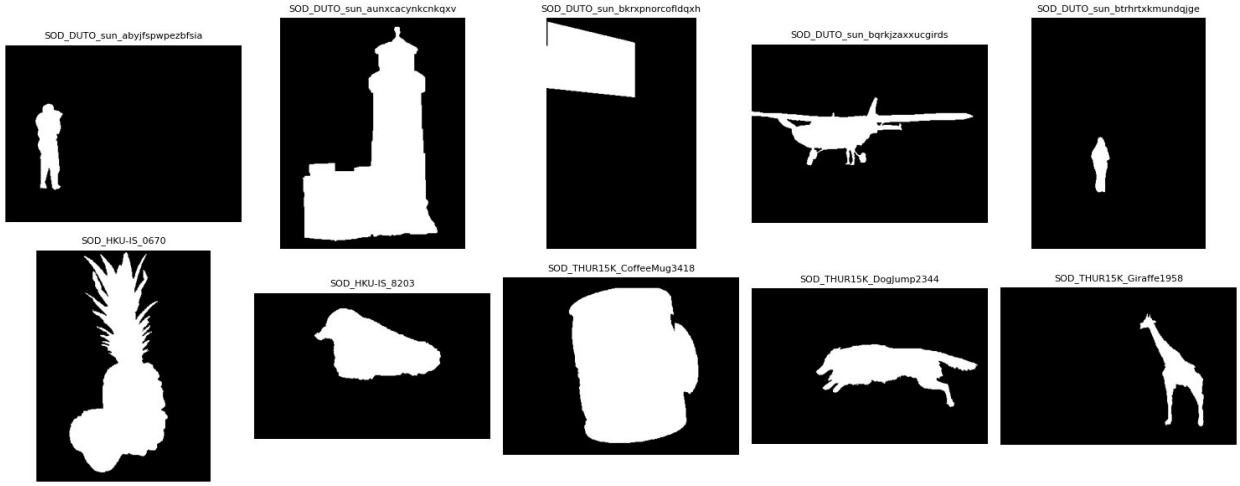
Object Mask and Envelope

The object mask identifies the exact region occupied by the camouflaged object, while the envelope represents the immediate surrounding region of the object. The envelope serves as the local background against which the camouflage is evaluated.

By comparing the object with its surrounding envelope, we can determine how well the generated camouflage blends with its local environment.



Masks used for sampling



Detector Confidence

A pre-trained camouflage object detector (SINet-V2) is used to obtain confidence scores for both the original and generated images.

The detector confidence indicates how likely the model believes that a camouflaged object is present.

- A lower confidence score after generation suggests that the object has become more difficult to detect, indicating improved camouflage.

- A higher confidence score suggests that the generated camouflage has made the object easier to detect.

```
display(original_df)

display(generated_df)
```

	image	mask	object_confidence	background_confidence	maximum_confidence	contrast
0	SOD_DUTO_sun_abyjfspwpezbfsia.jpg	SOD_DUTO_sun_abyjfspwpezbfsia.png	0.959801	0.021681	1.000000	0.938120
1	SOD_DUTO_sun_aunxcacynkcnkqxv.jpg	SOD_DUTO_sun_aunxcacynkcnkqxv.png	0.046772	0.000507	0.680960	0.046265
2	SOD_DUTO_sun_bkrxpnorcofldqxh.jpg	SOD_DUTO_sun_bkrxpnorcofldqxh.png	0.000562	0.026553	0.071395	-0.025991
3	SOD_DUTO_sun_bqrkjzaxxucgirds.jpg	SOD_DUTO_sun_bqrkjzaxxucgirds.png	0.489968	0.006097	1.000000	0.483871
4	SOD_DUTO_sun_btrhrtxkmundqjge.jpg	SOD_DUTO_sun_btrhrtxkmundqjge.png	0.833105	0.000752	1.000000	0.832353
5	SOD_HKU-IS_0670.jpg	SOD_HKU-IS_0670.png	0.003265	0.260764	0.989351	-0.257498
6	SOD_HKU-IS_8203.jpg	SOD_HKU-IS_8203.png	0.006998	0.895692	0.999696	-0.888694
7	SOD_THUR15K_CoffeeMug3418.jpg	SOD_THUR15K_CoffeeMug3418.png	0.012971	0.800563	0.999802	-0.787593
8	SOD_THUR15K_DogJump2344.jpg	SOD_THUR15K_DogJump2344.png	0.018206	0.979197	1.000000	-0.960992
9	SOD_THUR15K_Giraffe1958.jpg	SOD_THUR15K_Giraffe1958.png	0.072348	0.943333	1.000000	-0.870985

	image	mask	object_confidence	background_confidence	maximum_confidence	contrast
0	004_SOD_DUTO_sun_aunxcacynkcnkqxv_result.png	SOD_DUTO_sun_aunxcacynkcnkqxv.png	0.725027	0.026076	1.000000	0.698951
1	005_SOD_THUR15K_Giraffe1958_result.png	SOD_THUR15K_Giraffe1958.png	0.871211	0.076549	1.000000	0.794662
2	006_SOD_DUTO_sun_abyjfspwpezbfsia_result.png	SOD_DUTO_sun_abyjfspwpezbfsia.png	0.806324	0.043744	0.999998	0.762580
3	007_SOD_THUR15K_DogJump2344_result.png	SOD_THUR15K_DogJump2344.png	0.936659	0.013538	1.000000	0.923121
4	008_SOD_HKU-IS_8203_result.png	SOD_HKU-IS_8203.png	0.648117	0.073970	0.999932	0.574147
5	010_SOD_DUTO_sun_bqrkjzaxxucgirds_result.png	SOD_DUTO_sun_bqrkjzaxxucgirds.png	0.522259	0.011992	0.999997	0.510267
6	011_SOD_THUR15K_CoffeeMug3418_result.png	SOD_THUR15K_CoffeeMug3418.png	0.967648	0.013485	1.000000	0.954163
7	012_SOD_DUTO_sun_bkrxpnorcofldqxh_result.png	SOD_DUTO_sun_bkrxpnorcofldqxh.png	0.001996	0.031092	0.157378	-0.029095
8	013_SOD_DUTO_sun_btrhrtxkmundqjge_result.png	SOD_DUTO_sun_btrhrtxkmundqjge.png	0.903695	0.069900	0.999636	0.833795
9	014_SOD_HKU-IS_0670_result.png	SOD_HKU-IS_0670.png	0.691764	0.037159	1.000000	0.654605

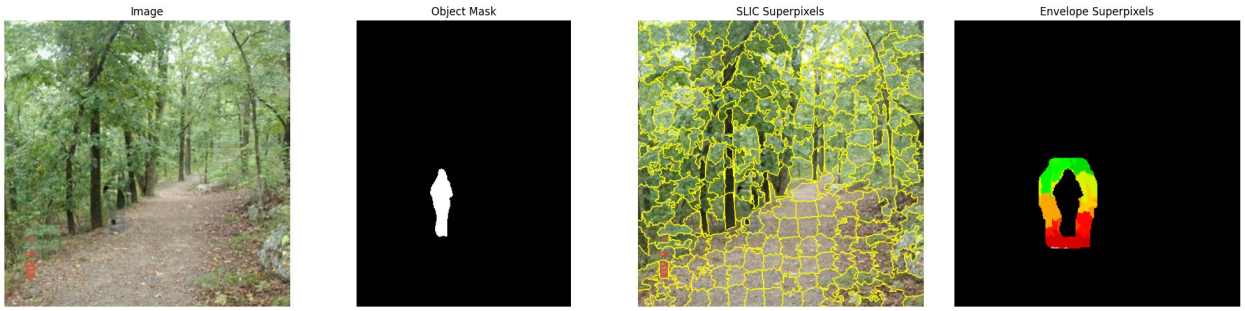
- Thus, detector confidence provides a quantitative measure of camouflage effectiveness.

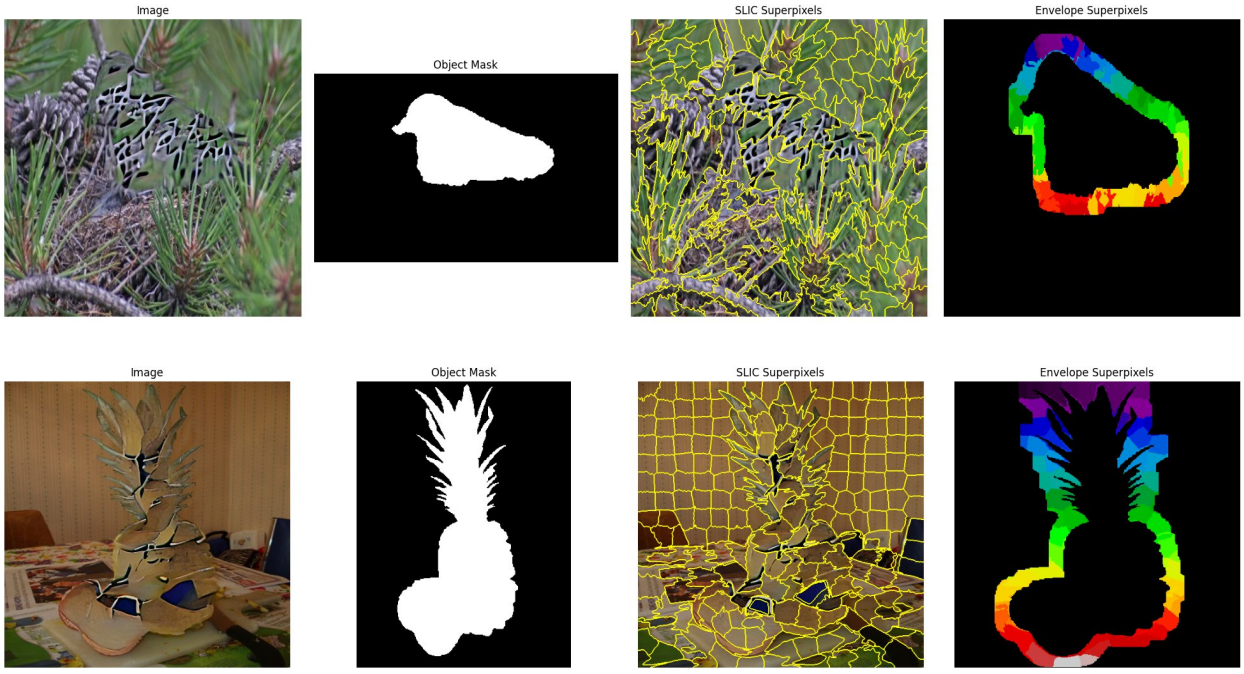
Superpixel Analysis

Superpixels partition the image into perceptually homogeneous regions while preserving important boundaries.Generating superpixels within the object's envelope allows us to analyze the local texture and structural consistency between the object and its surrounding environment.

A well-camouflaged object is expected to exhibit superpixel patterns that are visually consistent with those of the surrounding background.

Here i used SLIC (untrained fro the superpixel analysis.)





Histogram Analysis

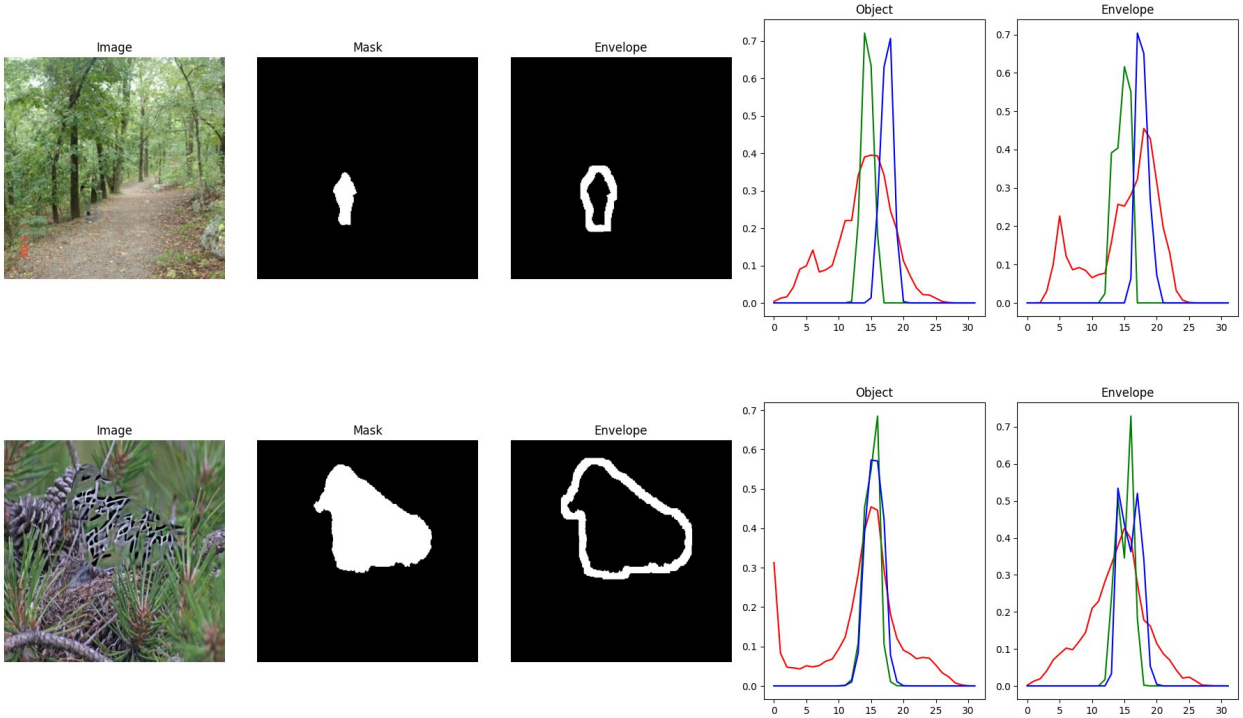
Color histograms are computed separately for the object region and the surrounding envelope.

These histograms describe the distribution of pixel intensities and colors within each region.

A successful camouflage should produce color distributions that closely resemble those of the surrounding environment.

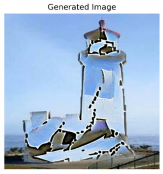
Histogram similarity metrics such as Bhattacharyya distance, Histogram Intersection, and Chi-Square distance are computed to quantify this similarity.

	image	L_Bhattacharyya	A_Bhattacharyya	B_Bhattacharyya	L_Intersection	A_Intersection	B_Intersection	L_ChiSquare	A_ChiSquare	B_ChiSquare
0	004_SOD_DUTO_sun_aunxcacynkcnkqxv_result.png	0.301747	0.102354	0.258598	2.935217	1.283018	1.760011	1.777088	0.052595	0.909761
1	005_SOD_THUR15K_Giraffe1958_result.png	0.366563	0.204034	0.122812	2.449228	1.257276	1.593920	99.106153	0.239083	0.396132
2	006_SOD_DUTO_sun_abyjfspwpezbfsia_result.png	0.172299	0.217603	0.244339	4.050110	1.003540	1.280781	0.927884	0.495157	2.407435
3	007_SOD_THUR15K_DogJump2344_result.png	0.225200	0.250573	0.245848	2.642860	0.994620	1.236029	0.934328	0.744421	0.750619
4	008_SOD_HKU-IS_8203_result.png	0.239879	0.129182	0.202263	3.218637	1.716559	1.735410	1.077352	0.305391	1.318974
5	010_SOD_DUTO_sun_bqrkjzaxxucgirds_result.png	0.322235	0.213537	0.305633	2.595982	1.137137	1.179819	2.082405	1.120941	3.407851
6	011_SOD_THUR15K_CoffeeMug3418_result.png	0.379269	0.275788	0.184300	2.437721	1.170094	1.306882	2.487993	0.471102	0.215719
7	012_SOD_DUTO_sun_bkrxpnorcofldqxh_result.png	0.327642	0.247742	0.192902	3.478534	1.513039	1.465400	2.348606	2.493431	0.474317
8	013_SOD_DUTO_sun_btrhrtxkmundqjge_result.png	0.234892	0.215108	0.194614	2.784099	1.420616	1.534409	1.930317	1.134415	1.524895
9	014_SOD_HKU-IS_0670_result.png	0.247880	0.282513	0.231131	2.373461	0.918856	1.265095	1.009010	0.965642	0.543660


Image
Mask
Envelope
Object
Envelope
Image
Mask
Envelope
Object
Envelope

Analysis report for sample images

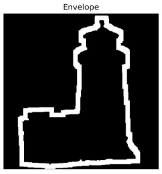
Generated Image



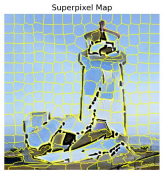
Object Mask



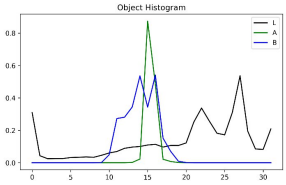
Envelope



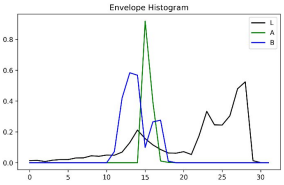
Superpixel Map



Object Histogram



Envelope Histogram



Object Pixels : 89841

Envelope Pixels : 28446

Object Confidence : 0.725

Background Confidence : 0.026

Contrast : 0.699

Max Confidence : 1.000

L Bhattacharyya : 0.302

A Bhattacharyya : 0.102

B Bhattacharyya : 0.259

L Intersection : 2.935

A Intersection : 1.283

B Intersection : 1.760

L ChiSquare : 1.777

A ChiSquare : 0.053

B ChiSquare : 0.910

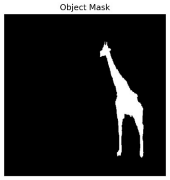
Mask Overlay



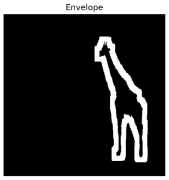
Generated Image



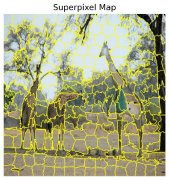
Object Mask



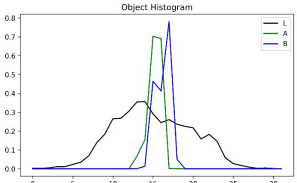
Envelope



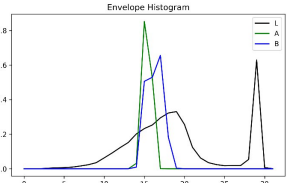
Superpixel Map



Object Histogram



Envelope Histogram



Object Pixels : 13233

Envelope Pixels : 20609

Object Confidence : 0.871

Background Confidence : 0.077

Contrast : 0.795

Max Confidence : 1.000

L Bhattacharyya : 0.367

A Bhattacharyya : 0.204

B Bhattacharyya : 0.123

L Intersection : 2.449

A Intersection : 1.257

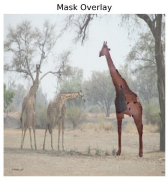
B Intersection : 1.594

L ChiSquare : 99.106

A ChiSquare : 0.239

B ChiSquare : 0.396

Mask Overlay



Camouflage Effectiveness Metric (CEM)

The objective of this work is to develop a Camouflage Effectiveness Metric (CEM) that quantitatively measures how well an object blends with its immediate surroundings. Unlike traditional evaluation methods that rely on the performance of camouflage object detection models, this metric evaluates camouflage directly from the visual characteristics of the object and its local background.

The underlying hypothesis is that a well-camouflaged object should exhibit high similarity with its surrounding environment in terms of color, texture, structure, boundaries, and semantic appearance.

The proposed metric compares the object with a morphological envelope, which represents the immediate local background surrounding the object. Multiple complementary similarity measures are extracted and combined to obtain a single camouflage effectiveness score.

Methodology

The complete pipeline consists of five major stages:

1. Image Preprocessing
2. Region Extraction
3. Feature Extraction
4. Similarity Measurement
5. Feature Fusion to obtain the final Camouflage Effectiveness Metric

1. Image Preprocessing

The notebook starts by loading the RGB image and its corresponding binary object mask.

The provided masks contain:

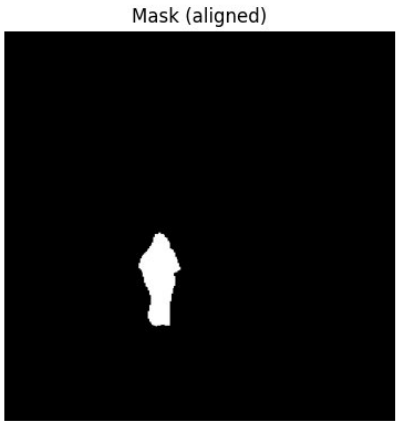
- White pixels representing the background
- Black pixels representing the object

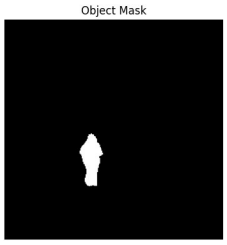
Since all subsequent operations require the object to be represented by white pixels, the masks are first binarized and then inverted.

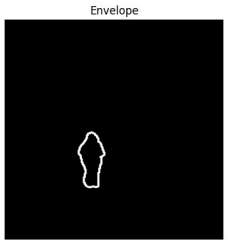
A **morphological dilation** is then applied to the object mask to generate a surrounding region called the **Envelope**.

Envelope = Dilated Mask − Object Mask











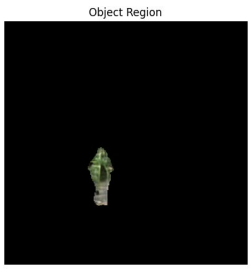
2. Region Extraction

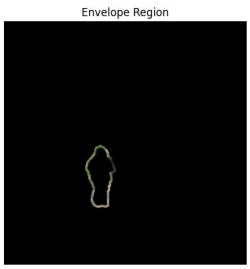
Using the binary mask and the generated envelope, two regions are extracted from the original RGB image:

- Object Region
- Envelope Region

These two regions form the basis for all subsequent feature comparisons.

Instead of comparing the object with the entire image, all similarity measurements are computed only between the object and its local surroundings. This makes the metric independent of global image statistics.





3. Feature Extraction

To capture different visual aspects of camouflage, multiple complementary features are extracted.

A. Color Distribution Similarity

The RGB image is converted into the HSV color space.

HSV is chosen because it separates

- Hue
- Saturation
- Intensity

making it more robust to illumination changes than RGB.

Normalized histograms are computed independently for

- Hue
- Saturation
- Value

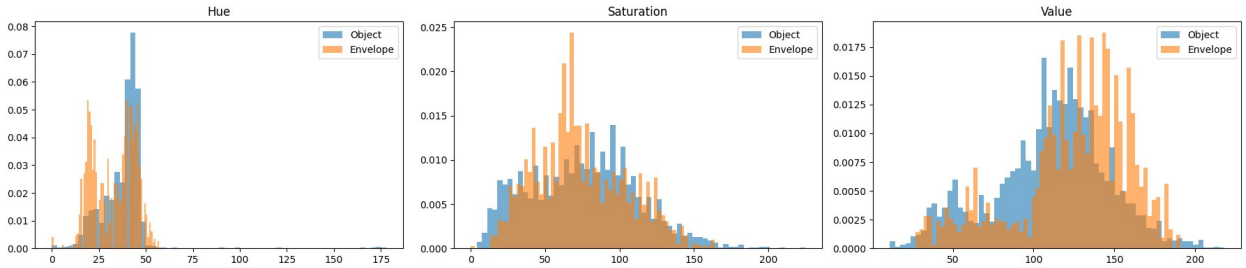
for both the object and the envelope.

Histogram Intersection is then used to measure the similarity between the corresponding histograms.

The weighted average of the three channels represents the overall color similarity.

Purpose

Measures how similar the color distributions of the object and its surroundings are.



B. Texture Similarity

Texture similarity is measured using **Local Binary Patterns (LBP)**.

The image is first converted into grayscale.

Uniform LBP descriptors are computed for the entire image.

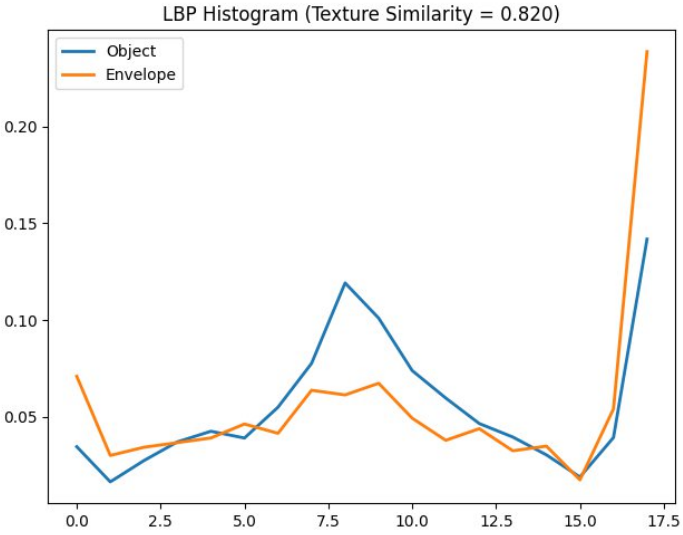
LBP histograms are then generated separately for

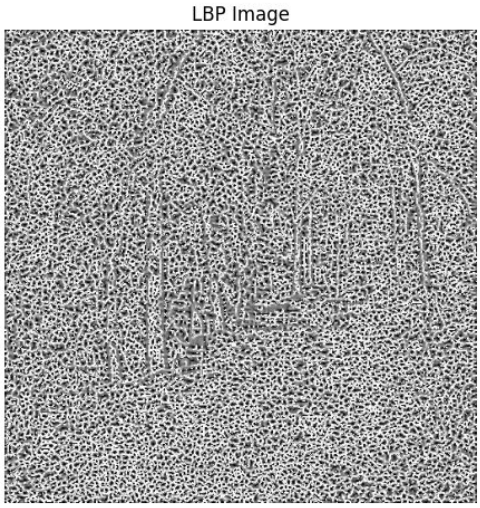
- Object
- Envelope

Histogram Intersection is used to compare these distributions.

Purpose

Measures whether the object exhibits texture patterns similar to the surrounding environment.





C. Superpixel Consistency

The image is segmented into homogeneous regions using the **SLIC Superpixel Algorithm**.

Each superpixel is represented by

- Mean HSV values
- Mean LBP value

Superpixels are classified as

- Object
- Envelope

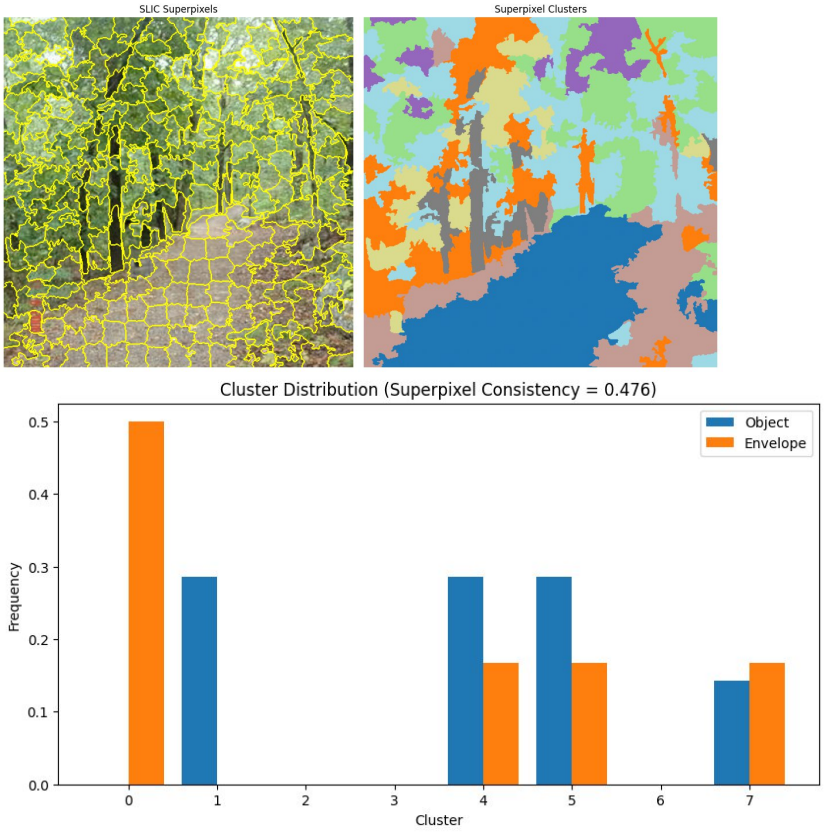
using a purity threshold.

The resulting feature vectors are clustered using K-Means.

Finally, the cluster distributions of object and envelope superpixels are compared using Histogram Intersection.

Purpose

Measures whether the object is composed of similar visual regions as its surrounding background.



D. Boundary Similarity

Human observers often detect camouflaged objects by identifying discontinuities along object boundaries.

To quantify this effect,

inner and outer boundary bands are generated using morphological erosion and dilation.

Within these boundary regions,

HSV histograms are computed and compared.

Gradient histograms along the boundaries are also compared.

The two measures are combined to obtain the final Boundary Similarity Score.

Purpose

Measures how smoothly the object transitions into the surrounding environment.



E. Feature Embedding Similarity

High-level semantic similarity is extracted using a pretrained **ResNet-50** feature extractor.

The classification layer is removed and only the feature extraction backbone is retained.

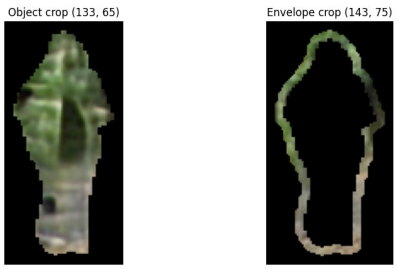
Feature vectors are independently extracted for

- Object Region
- Envelope Region

Cosine Similarity between the two feature vectors provides the embedding similarity score.

Purpose

Measures semantic similarity beyond low-level appearance.



F. Gradient Similarity

Structural similarity is measured using the **Histogram of Oriented Gradients (HOG)**.

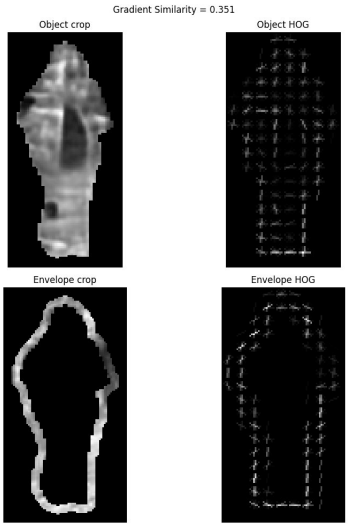
HOG descriptors are computed independently for

- Object
- Envelope

Cosine Similarity between the descriptors provides the gradient similarity score.

Purpose

Measures similarity in edge orientation and structural appearance.



4. Similarity Measures

The notebook computes six independent similarity scores.

Feature	Similarity Measure

Color Distribution	Histogram Intersection
Texture	Histogram Intersection
Superpixel Consistency	Histogram Intersection
Boundary Similarity	Histogram Intersection + Gradient Similarity
Deep Feature Similarity	Cosine Similarity
Gradient Similarity	Cosine Similarity

All feature scores are constrained to the range **[0,1]**, where

- **1** indicates high similarity between object and envelope.
- **0** indicates low similarity.

5. Camouflage Effectiveness Metric (CEM)

The final camouflage score is computed as a weighted combination of the six similarity measures.

The current implementation uses the following weights:

Feature	Weight
Color Similarity	0.20
Texture Similarity	0.15

Superpixel Consistency	0.15
Boundary Similarity	0.20
Feature Embedding Similarity	0.20
Gradient Similarity	0.10

The final score is therefore computed as

CEM =

0.20 × Color Similarity

+ 0.15 × Texture Similarity

+ 0.15 × Superpixel Consistency

+ 0.20 × Boundary Similarity

+ 0.20 × Embedding Similarity

+ 0.10 × Gradient Similarity

Higher scores indicate that the object is more visually similar to its immediate surroundings and is therefore expected to exhibit stronger camouflage.

Expected Outcome

This notebook produces

- Object and envelope extraction
- HSV color similarity
- LBP texture similarity
- Superpixel consistency analysis
- Boundary similarity analysis
- Deep feature embedding similarity
- HOG gradient similarity

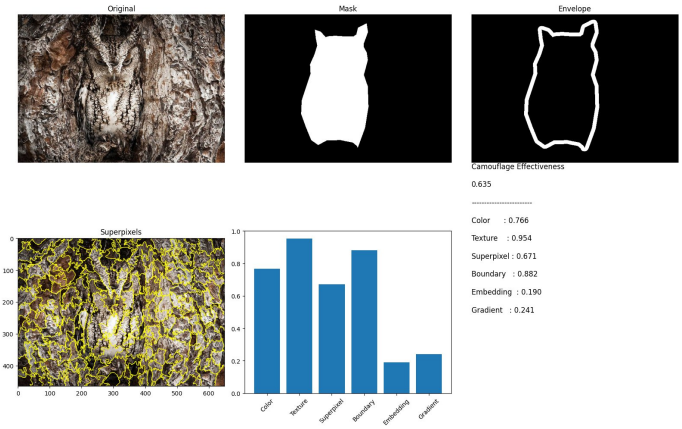
- Individual feature scores
- Final Camouflage Effectiveness Metric (CEM)

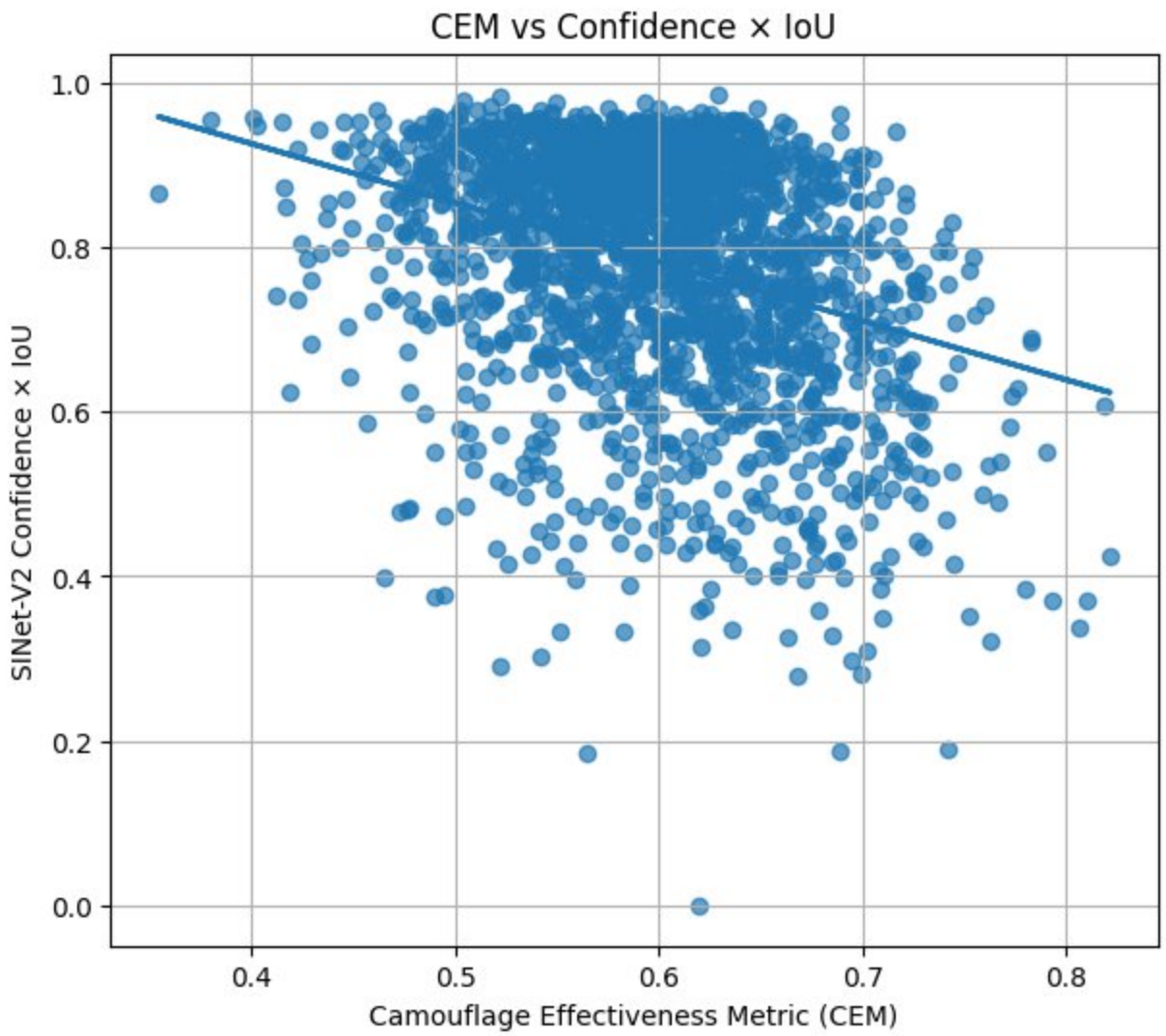
The proposed metric is designed to be **detector-independent**, allowing camouflage quality to be evaluated directly from the image without relying on the performance of a specific camouflage object detection mod

What does it tells

- Average camouflage difficulty.
- Distribution of easy vs. hard images.
- Which visual cues (color, texture, edges, semantics) contribute to camouflage.
- Which images are most challenging.
- Whether camouflage genuinely affects detector confidence and segmentation accuracy.
- Whether the detector behaves consistently across the dataset

Overall conclusion





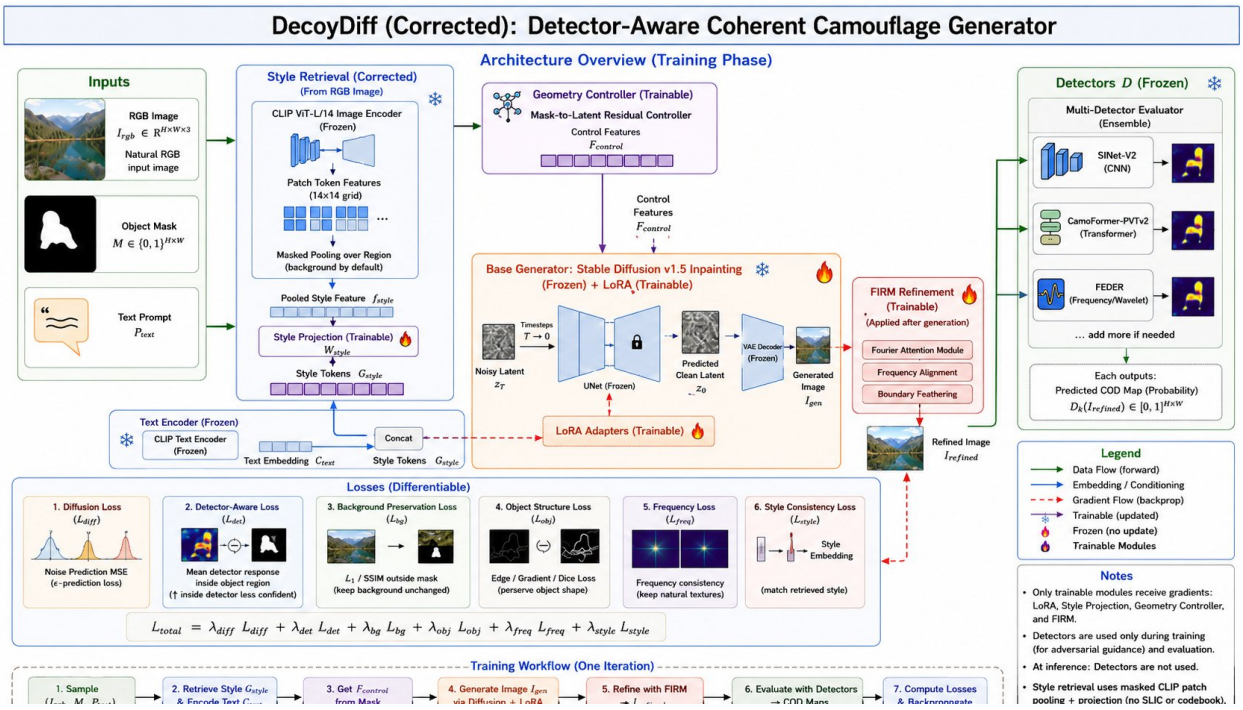
Feature	n	Pearson r (vs Confidence*IoU)	Pearson p	Spearman rho (vs Confidence*IoU)	Spearman p
Color	2000	-0.021284	3.414182e-01	-0.017251	4.406808e-01
Texture	2000	-0.020071	3.696473e-01	0.001591	9.433038e-01
Superpixel	2000	-0.038450	8.559396e-02	-0.058573	8.790722e-03
Boundary	2000	0.097917	1.150470e-05	0.065098	3.585207e-03
Embedding	2000	-0.626830	8.306490e-219	-0.613893	1.780913e-207
Gradient	2000	-0.089905	5.665310e-05	-0.048590	2.978396e-02

```
============================================================
CEM vs IoU
Pearson  : r = -0.3135, p = 7.4405e-47
Spearman : ρ = -0.3073, p = 5.2661e-45
============================================================
CEM vs Confidence
Pearson  : r = -0.3757, p = 4.7139e-68
Spearman : ρ = -0.3302, p = 4.4429e-52
============================================================
CEM vs Confidence*IoU
Pearson  : r = -0.3279, p = 2.3282e-51
Spearman : ρ = -0.3135, p = 7.4597e-47
============================================================
Confidence vs IoU
Pearson  : r = 0.8862, p = 0.0000e+00
Spearman : ρ = 0.9378, p = 0.0000e+00
```

Our notebook(metric) effectively performs **three levels of analysis**:

1. **Image-level analysis:** It quantifies the camouflage characteristics of each image using multiple complementary visual features (color, texture, boundary, gradient, superpixels, and deep semantic embeddings) and combines them into a unified Camouflage Effectiveness Metric (CEM).
2. **Dataset-level analysis:** By aggregating CEM across all images, it characterizes the overall difficulty, variability, and distribution of camouflage within the dataset, enabling the identification of easy, difficult, and outlier samples.
3. **Performance-level analysis:** By correlating CEM with detector confidence and IoU, it evaluates whether increased visual camouflage translates into decreased detection performance. Your results indicate that this relationship exists, is statistically significant, and follows the expected direction (higher camouflage leads to lower confidence and lower IoU), while also showing that camouflage alone does not fully explain detector performance, suggesting that additional factors influence segmentation accuracy.

Proposed Method: DecoyDiff — Detector-Aware Coherent Camouflage Generator



In this work, I propose **DecoyDiff**, a detector-aware camouflage image generation framework. The aim of this model is to generate visually realistic camouflage images while reducing the confidence of camouflaged object detection models. Unlike standard image inpainting methods, which mainly focus on filling a masked region realistically, DecoyDiff also considers whether the generated object can confuse or suppress multiple frozen camouflage detectors.

The proposed model takes three inputs: an RGB image, a binary object mask, and a text prompt. The RGB image is represented as

$$I_{rgb} \in [0,1]^{H \times W \times 3}$$

where H and W represent image height and width. The binary object mask is represented as

$$M \in \{0,1\}^{H \times W}$$

where $M = 1$ denotes the object region to be camouflaged and $M = 0$ denotes the background region. The text prompt is represented as

$$P_{text}$$

The goal of the model is to generate a refined camouflage image

$$I_{refined} \in [0,1]^{H \times W \times 3}$$

such that the masked object region becomes visually coherent with the surrounding background, while the background outside the mask remains unchanged. At the same time, the generated image should reduce the response of multiple frozen camouflaged object detectors.

Objective

The proposed model can be formulated as a conditional image generation problem. Given an input image I_{rgb}, object mask M, and prompt P_{text}, the generator produces a camouflaged image:

$$I_{refined} = G(I_{rgb}, M, P_{text})$$

The main objective is to optimize the generator so that the output image satisfies three conditions:

$$I_{refined} \approx I_{rgb} \quad \text{outside the mask}$$

$$I_{refined} \text{ is visually realistic and texture-consistent}$$

$$D_k(I_{refined}) \to 0 \quad \text{inside the object mask}$$

where D_k denotes the kth frozen camouflage detector. The detectors are not trained; they are only used to provide detector-aware gradients during training.

The trainable parameters of DecoyDiff are

$$\Theta = \{\theta_{LoRA}, \theta_{ControlNeXt}, \theta_{FIRM}\}$$

where θ_{LoRA} represents the LoRA parameters added to Stable Diffusion, $\theta_{ControlNeXt}$ represents the trainable geometry controller, and θ_{FIRM} represents the frequency refinement module. The Stable Diffusion backbone, VAE, CLIP encoders, and camouflage detectors are kept frozen.

Style retrieval block

The current DecoyDiff pipeline uses frozen CLIP vision features pooled over the masked region, projects them into trainable style tokens, and concatenates those with frozen text embeddings before conditioning the frozen Stable Diffusion inpainting UNet with trainable LoRA. The generated image is then refined by FIRM, a trainable frequency-domain refinement module that combines Fourier attention with boundary feathering, and a frozen detector ensemble provides detector-aware loss during training only."

To generate camouflage that is visually compatible with the input image, I use a style retrieval block. This block extracts local style information from the RGB image using a frozen CLIP image encoder.

First, the input image is passed through the CLIP image encoder:

$$F_{clip} = E_{img}(I_{rgb})$$

where E_{img} is the frozen CLIP image encoder and F_{clip} is the extracted visual feature map.

To focus on the object region, masked average pooling is applied using the object mask:

$$c_{query} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W} M_{ij} F_{clip}(i,j)}{\sum_{i=1}^{H}\sum_{j=1}^{W} M_{ij} + \epsilon}$$

where ϵ is a small constant used to avoid division by zero. This gives a localized style query vector c_{query}.

Then, this query vector is compared with a visual style codebook $\mathcal{C}$. The most similar style embedding is retrieved using cosine similarity:

$$c_{style} = \arg\max_{c_i \in \mathcal{C}} \frac{c_{query} \cdot c_i}{\|c_{query}\|_2 \|c_i\|_2}$$

Here, c_{style} represents the retrieved style embedding. This embedding helps the generator produce textures that are consistent with the surrounding image.

Text Conditioning

The text prompt P_{text} is encoded using a frozen CLIP text encoder:

$$c_{text} = E_{text}(P_{text})$$

where E_{text} is the CLIP text encoder and c_{text} is the text embedding. The diffusion generator is conditioned using both the text embedding and the retrieved style embedding:

$$c = \{c_{text}, c_{style}\}$$

This allows the model to use semantic information from the prompt and visual style information from the image.

Latent Diffusion Inpainting

The base generator of DecoyDiff is Stable Diffusion Inpainting. Instead of generating the image directly ir pixel space, the image is first encoded into latent space using a frozen VAE encoder:

$$z_0 = E_{VAE}(I_{rgb})$$

During training, Gaussian noise is added to the clean latent z_0 at timestep t:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

where

$$\epsilon \sim \mathcal{N}(0, I)$$

and $\bar{\alpha}_t$ is the cumulative noise scheduling coefficient.

The denoising UNet predicts the added noise:

$$\hat{\epsilon} = \epsilon_\theta(z_t, t, c_{text}, c_{style}, M, F_{control})$$

where $F_{control}$ represents the spatial control features produced by the ControlNeXt module.

The diffusion loss is defined as the mean squared error between the true noise and the predicted noise:

$$L_{diff} = \mathbb{E}_{z_0,t,\epsilon}\left[\|\epsilon - \hat{\epsilon}\|_2^2\right]$$

This loss encourages the diffusion model to generate realistic content in the masked region.

LoRA-Based Fine-Tuning

Instead of fine-tuning the full Stable Diffusion model, I use Low-Rank Adaptation, or LoRA. LoRA updates only small low-rank matrices while keeping the original model weights frozen.

For a frozen weight matrix W, LoRA modifies it as:

$$W' = W + \Delta W$$

where

$$\Delta W = \frac{\alpha}{r} BA$$

Here, A and B are trainable low-rank matrices:

$$A \in \mathbb{R}^{r \times d_{in}}, \qquad B \in \mathbb{R}^{d_{out} \times r}$$

where r is the LoRA rank and α is a scaling factor. This makes the training more efficient because only a small number of parameters are updated.

ControlNeXt Geometry Controller

The ControlNeXt geometry controller is used to preserve object geometry and spatial structure. It takes the object mask and mask boundary information as input and produces control features:

$$F_{control} = G_{\psi}(M, \nabla M)$$

where G_{ψ} is the trainable ControlNeXt module and ∇M represents the gradient or edge information of the mask.

These control features are injected into the Stable Diffusion UNet. At a UNet layer l, the feature injection can be represented as:

$$h'_l = h_l + \gamma_l A_l(F^l_{control})$$

where h_l is the original UNet feature map, $F^l_{control}$ is the control feature at the same resolution, A_l is a projection layer, and γ_l controls the strength of feature injection.

This helps the diffusion model generate camouflage while respecting the object location and mask structure.

Image Decoding and Masked Composition

After the denoising process, the predicted clean latent is decoded back into image space using the frozen VAE decoder:

$$\hat{I} = D_{VAE}(\hat{z}_0)$$

To ensure that the background remains unchanged, the generated image is composed with the original image using the binary mask:

$$I_{gen} = M \odot \hat{I} + (1 - M) \odot I_{rgb}$$

where $\odot$ denotes element-wise multiplication. This operation ensures that the generated content is mainly applied inside the masked object region, while the original background is preserved.

Frequency Refinement Module

After the diffusion model generates the initial camouflaged image, I use a Frequency Refinement Module, or FIRM, to improve texture consistency and reduce visible artifacts near the mask boundary.

For an image I, the two-dimensional Fourier transform can be written as:

$$\mathcal{F}(I) = A(I)e^{j\Phi(I)}$$

where $A(I)$ is the amplitude spectrum and $\Phi(I)$ is the phase spectrum. The amplitude mainly represents texture and frequency distribution, while the phase mainly preserves spatial structure.

The FIRM module refines the generated image as:

$$I_{firm} = R_\phi(I_{gen}, M)$$

where R_ϕ is the trainable refinement module.

To align the texture of the generated object region with the surrounding background, a frequency consistency loss is used:

$$L_{freq} = \|A(I_{firm} \odot M) - A(I_{rgb} \odot B(M))\|_1$$

where $B(M)$ denotes a local background band around the object mask. This helps the generated object region follow the frequency characteristics of the nearby background.

To avoid sharp transitions at the object boundary, a softened mask $\tilde{M}$ is used for boundary feathering:

$$I_{refined} = \tilde{M} \odot I_{firm} + (1 - \tilde{M}) \odot I_{rgb}$$

where $\tilde{M}$ is obtained by smoothing or blurring the binary mask M. This produces a smoother transition between the generated region and the original background.

Detector-Aware Loss

A key part of DecoyDiff is the detector-aware loss. I use multiple frozen camouflaged object detectors during training. Let the detector ensemble be

$$D = \{D_1, D_2, \ldots, D_K\}$$

Each detector predicts a probability map:

$$P_k = D_k(I_{refined})$$

where

$$P_k \in [0,1]^{H \times W}$$

The detector-aware loss is computed as the average detector response inside the object mask:

$$L_{det} = \frac{1}{K} \sum_{k=1}^{K} \frac{\sum_{i=1}^{H} \sum_{j=1}^{W} M_{ij} P_k(i,j)}{\sum_{i=1}^{H} \sum_{j=1}^{W} M_{ij} + \epsilon}$$

By minimizing this loss, the generator learns to reduce the detector confidence in the camouflaged object region.

The detectors are frozen, so their parameters are not updated:

$$\nabla_{\theta_{D_k}} L_{det} = 0$$

However, gradients are still propagated through the detector outputs back to the generator:

$$\nabla_{\Theta} L_{det} = \frac{\partial L_{det}}{\partial I_{refined}} \frac{\partial I_{refined}}{\partial \Theta}$$

This allows the generator to learn detector-suppressing camouflage patterns without modifying the detectors themselves.

Background Preservation Loss

To prevent unnecessary changes outside the mask, I use a background preservation loss. This loss penalizes differences between the original image and the refined image in the background region:

$$L_{bg} = \frac{\|(1 - M) \odot (I_{refined} - I_{rgb})\|_1}{\sum_{i=1}^{H} \sum_{j=1}^{W} (1 - M_{ij}) + \epsilon}$$

An SSIM-based term can also be added to preserve structural similarity:

$$L_{bg}^{ssim} = 1 - SSIM((1 - M) \odot I_{refined}, (1 - M) \odot I_{rgb})$$

Therefore, the complete background preservation loss is:

$$L_{bg}^{total} = L_{bg} + \lambda_{ssim} L_{bg}^{ssim}$$

This ensures that the model focuses on modifying the object region and does not unnecessarily distort the background.

Object Structure Loss

Although the object must become camouflaged, its spatial structure should not be completely destroyed. To preserve object shape and boundary consistency, I use an object structure loss.

The boundary of the mask is extracted using image gradients:

$$B(M) = |\nabla_x M| + |\nabla_y M|$$

Similarly, the edge map of the refined image can be obtained as:

$$E(I_{refined}) = |\nabla_x I_{refined}| + |\nabla_y I_{refined}|$$

The object structure loss is defined as:

$$L_{obj} = \|B(M) \odot E(I_{refined}) - B(M) \odot E(I_{rgb})\|_1$$

This encourages the generated image to maintain reasonable object boundary information while still blending with the background.

A Dice-style boundary term can also be used:

$$L_{dice} = 1 - \frac{2\sum B(M)E(I_{refined}) + \epsilon}{\sum B(M) + \sum E(I_{refined}) + \epsilon}$$

The complete object structure loss is:

$$L_{obj}^{total} = L_{obj} + \lambda_{dice} L_{dice}$$

Style Consistency Loss

To ensure that the generated camouflage follows the retrieved style, a style consistency loss is used. The refined masked region is encoded using the frozen CLIP image encoder:

$$c_{gen} = E_{img}(I_{refined} \odot M)$$

The generated style embedding is then compared with the retrieved style embedding c_{style} using cosine distance:

$$L_{style} = 1 - \frac{c_{gen} \cdot c_{style}}{\|c_{gen}\|_2 \|c_{style}\|_2}$$

Minimizing this loss encourages the generated object region to match the visual style retrieved from the input image or the style codebook.

Final Training Objective

The final training objective combines all the losses:

$$L_{total} = \lambda_{diff} L_{diff} + \lambda_{det} L_{det} + \lambda_{bg} L_{bg}^{total} + \lambda_{obj} L_{obj}^{total} + \lambda_{freq} L_{freq} + \lambda_{style} L_{style}$$

where each λ controls the contribution of the corresponding loss term.

The full optimization problem is written as:

$$\Theta^* = \arg\min_{\Theta} \mathbb{E}_{(I_{rgb}, M, P_{text}) \sim \mathcal{D}} \left[L_{total} \right]$$

where

$$\Theta = \{\theta_{LoRA}, \theta_{ControlNeXt}, \theta_{FIRM}\}$$

Only these parameters are updated during training. The Stable Diffusion backbone, CLIP encoders, VAE, and camouflaged object detectors remain frozen.

Inference Phase

During inference, the detector ensemble is not required. The detectors are used only during training to provide adversarial supervision. Given an input image, mask, and prompt, the trained generator produces the final camouflaged image as:

$$I_{out} = R_{\phi} \left(G_{\theta}(I_{rgb}, M, P_{text}, c_{style}, F_{control}) \right)$$

This makes inference more efficient because only the trained generator, ControlNeXt module, style retrieval block, and FIRM module are needed.

Training Algorithm

The training process for one iteration is as follows.

First, an image, object mask, and text prompt are sampled from the dataset:

$$(I_{rgb}, M, P_{text}) \sim \mathcal{D}$$

The text prompt is encoded using the frozen CLIP text encoder:

$$c_{text} = E_{text}(P_{text})$$

The style embedding is retrieved using the frozen CLIP image encoder and visual codebook:

$$c_{style} = R(E_{img}(I_{rgb}), M)$$

The ControlNeXt module generates spatial control features from the mask:

$$F_{control} = G_{\psi}(M, \nabla M)$$

The image is encoded into latent space:

$$z_0 = E_{VAE}(I_{rgb})$$

A timestep t and Gaussian noise ϵ are sampled:

$$t \sim \mathcal{U}(1, T), \qquad \epsilon \sim \mathcal{N}(0, I)$$

The noisy latent is then created:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

The LoRA-adapted UNet predicts the noise:

$$\hat{\epsilon} = \epsilon_{\theta}(z_t, t, c_{text}, c_{style}, M, F_{control})$$

The predicted latent is decoded to obtain the generated image, which is then refined by the FIRM module:

$$I_{gen} = D_{VAE}(\hat{z}_0)$$

$$I_{refined} = R_{\phi}(I_{gen}, M)$$

The refined image is passed through the frozen detector ensemble:

$$P_k = D_k(I_{refined}), \qquad k = 1, 2, \ldots, K$$

Finally, all losses are computed and combined:

$$L_{total} = \lambda_{diff} L_{diff} + \lambda_{det} L_{det} + \lambda_{bg} L_{bg}^{total} + \lambda_{obj} L_{obj}^{total} + \lambda_{freq} L_{freq} + \lambda_{style} L_{style}$$

The trainable parameters are updated by backpropagation:

$$\Theta \leftarrow \Theta - \eta \nabla_{\Theta} L_{total}$$

where η is the learning rate.

https://github.com/chaofengc/Awesome-Image-Quality-Assessment

https://github.com/dingkeyan93/DISTS